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Blockchain-enhanced SDN framework with deep unfolding spatial frequency attention network and pine cone optimization for smart city IoT cybersecurity

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Abstract

The rapid expansion of the Internet of Things (IoT) in smart cities has revolutionized services by connecting numerous devices to deliver intelligent applications. However, this interconnected environment also exposes IoT networks to severe cybersecurity risks, including data breaches, intrusions, and advanced persistent threats. Traditional detection mechanisms often struggle with high-dimensional traffic, noisy data, missing values, and evolving attack patterns, which significantly reduce their reliability. Moreover, ensuring secure and tamper-proof storage of detection outcomes in large-scale IoT system s remains a major challenge. To address these issues, this paper proposes a blockchain-enhanced IoT cybersecurity framework for smart cities, utilizing a software-defined networking framework with a Deep Unfolding Spatial-Frequency Attention Network and Pine Cone Optimization (DUSFAN-PCO). Initially, input data is collected from UNSW-NB15 and CIC-IDS2017 datasets. The collected data are pre-processed using Diffusion Models for Missing Value Imputation (DMMVI) to handle noise and missing values. Next, Carpet Weaver Optimization (CWO) is applied to select the most informative attributes. These selected features are then fed into the Deep Unfolding Spatial-Frequency Attention Network (DUSFAN), which classifies IoT network traffic into normal and multiple attack categories through real-time anomaly detection. To further enhance detection accuracy, Pine Cone Optimization (PCO) is used to fine-tune DUSFAN's weight parameters. Finally, the results are securely stored and managed using a Blockchain-Based Secure PBFT Lightweight Consensus (SPBFT-LC), ensuring immutability, transparency, and trust among IoT nodes. The proposed DUSFAN-PCO framework achieves an accuracy of 99.20% on UNSW-NB15 and 99.48% on CIC-IDS2017, with F1-scores of 99.52% and 99.50%, and ROC values of 0.99–1.00 across both datasets. Coupled with high blockchain-based security integrity (>99%), it ensures robust, real-time, and reliable detection of IoT network intrusions, making it highly suitable for secure and resilient smart city IoT environments.

Keywords Carpet weaver optimization · Deep unfolding spatial-frequency attention network · Diffusion models for missing value imputation · Internet of Things · Secure PBFT lightweight consensus

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Introduction

The IoT has increased manifold to transform contemporary city landscapes into smart cities, which are dependent on networking devices and smart services[1]. These networks are characterized by sensors, actuators, and communication technologies that improve transportation, energy, healthcare and governance[2]. This integration is also coupled with an extreme expansion of the quantity and sensitivity of information that ought to be moved across networks[3]. This information is supposed to be safeguarded to bring about trust, privacy and functionality. The cybersecurity concern has therefore become a significant issue in smart cities which are enabled by the IoT[4]. This is because the IoT environments are dynamic and heterogeneous and cannot be adequately dealt with by conventional network security mechanisms[5]. Denial of service, spoofing and data leakage have probabilities of harming essential services and creating enormous economic and social consequences[6]. The smart infrastructure of the city is too complicated to be covered with the same security solutions which are not scalable and adaptive[7]. Software-Defined Networking (SDN)[8] may be viewed as a possible solution to the problems. SDN provides centralization of control, programmability and flexibility to the network control by decoupling the control plane and the data plane[9]. In this facility, it is possible to monitor and track, analyze the traffic, and react immediately to the emergent threats[10]. SDN can also be used to allocate resources and dynamically add policies to different devices in the IoT[11]. The properties provide more active and dynamic cybersecurity operation in smart cities[12]. SDN and IoT would be integrated to offer increased visibility, coordination, and resilience to sophisticated attacks[13]. Besides, it is the starting point of the application of advanced methods, such as Machine Learning (ML), blockchain, and optimization algorithms[14]. With the emergence of smart cities, a sophisticated IoT cybersecurity turns out a root of secure, robust, and resilient cities[15].To address these needs, this study proposes a blockchain-enhanced DUSFAN-PCO framework that delivers intelligent, adaptive, and robust cybersecurity solutions tailored for IoT-driven smart city environments.

Novelty and Contribution,

The novelty of this work must be considered from the perspective of systems-level integration rather than in the context of each algorithm individually. The DMMVI, CWO, DUSFAN, PCO and SPBFT-LC algorithms are borrowed from or inspired by prior works. The innovation of this work lies in the integration of the above in one single architecture for SDN enabled IoT intrusions detection system and the associated analysis of the data reconstruction, adaptive feature selection, spatial-frequency deep learning, parameter optimization and secure blockchain-based events storage.

- Collects IoT network data from UNSW-NB15 and CIC-IDS2017 datasets and applies DMMVI to handle noise and missing values, ensuring high-quality input for analysis.
- Uses CWO to select the most informative features, reducing data dimensionality.
- Implements the DUSFAN for accurate real-time classification of normal and multiple attack types in IoT traffic.
- Applies PCO to fine-tune DUSFAN's weight parameters for improved precision and adaptability to evolving cyber threats.
- Provides a robust mechanism for real-time identification of anomalies in IoT network traffic, crucial for smart city environments
- Integrates a blockchain-based SPBFT-LC to ensure immutability, transparency, and trust among IoT nodes.
- Delivers an end-to-end secure, robust, and scalable intrusion detection solution for IoT networks in smart cities, combining data preprocessing, feature optimization, Deep Learning (DL), and blockchain storage.

In "Literature survey" section, pertinent literature is thoroughly evaluated. "Proposed methodology" section offers a thorough description of the methods employed in this research. The outcome and its discussion are examined in "Results and discussion" section. "Conclusion" section Conclusions with individual perspectives and suggestions for further investigation.

Literature survey

The most recent research on intelligent frameworks for enhancing IoT cybersecurity in smart cities is presented below.

In 2024, More et al.[16] have introduced a Machine Learning (ML)-based solution to improve the performance of Intrusion Detection Systems (IDSs) in IoT and smart network settings. Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), and decision tree algorithms were implemented utilizing the UNSW-NB15 network traffic dataset. To determine model performance, an exploratory data analysis was conducted, and a correlation analysis, along with random sampling, was used to select features. The findings suggest that the RM model proved to be most successful at identifying cyber-attacks and reinforcing IDS security.

In 2022, Tareq et al.[17] have introduced a DL-based solution to address the cybersecurity issue in IoT, and found that the attack was detected using multi-class classification. DenseNet and Inception Time are two intelligent network models, which were trained and tested on a massive scale of experiments. With the help of sliding window sequences and feature extraction, the models were able to detect many forms of cyber-attacks, thus making them to improve the security of the IoT.

In 2024, Priyadarshini et al.[18] have offered a new solution to cybersecurity improvement in smart cities, detecting the anomalies created by cyberattacks and guaranteeing privacy of data. The algorithm makes use of Federated Learning (FL) to learn models on distributed sources of data without raw data access. Classical ML and DL algorithm efficiency were compared, and it was found that privacy aware ones could be effectively deployed even to detect cyber threats, besides the ability to deploy scalable and resilient security policies in city IoT systems.

In 2024, Mishra et al.[19] suggested a hybrid model consisting of LSTM-SVM with min–max normalization, weighted average filtering, reptile search-based feature selection and Blockchain to ensure secured IoT transactions. This framework enhanced accuracy in detecting attacks and also assured security in smart city IoT communications.

In 2025, Vikram et al.[20] have come up with a blockchain architecture for securing IIoT, which involves the use of Blowfish encryption technique, Adaptive Wild Horse Optimization for key generation, and Capsule Bi-GRU attention mechanism for detecting attacks.

In 2025, This framework was put forward by Mahdi et al.[21]. This framework includes features of incremental learning, hybrid encryption, and blockchain. Through this model, the system keeps updating itself with the help of new data, stores data in the blockchain, and helps improve intrusion detection accuracy.

In 2024, Model suggested by Kaliyaperumal et al.[22] uses Deep Autoencoder, OCSVM and DBSCAN for DoS, DDoS and botnet attacks along with others as an Unsupervised Intrusion Detection Model. This model successfully deals with imbalance data and recognizes new threats.

In 2022, Awotunde et al.[23] presented a hybrid AI-based intrusion detection system for the security of the Internet of Things. The system includes steps such as data preprocessing and feature extraction, which increases the accuracy in attack detection. Through experiments, it was proven that it outperforms other systems in this field. In 2023, Saini et al.[24] developed a hybrid ensemble approach using techniques such as random forest, XGBoost, decision tree, CNN, and MLP to detect APT attacks in IoT networks.

In 2023, Abbas et al.[25] suggested a Federated Deep Neural Network (FDNN) framework for IoT cybersecurity that allows collaborative intrusion detection with high accuracy while maintaining the privacy of data using feature normalization, class balancing, and federated learning techniques. In recent research, attention-based deep learning, federated learning, and blockchain techniques have been used to increase the efficiency of IoT intrusion detection. Attention-based techniques make feature learning better and improve intrusion detection performance, whereas federated learning technique makes possible to train the model in a collaborative way without compromising on the privacy of users. In addition, blockchain-based intrusion detection system helps in secure and transparent storage of security events. The recent work done by Albanbay et al.[35], Alharthi et al.[36] and Kumar et al.[37] on federated learning and blockchain technique has proved their usefulness in providing efficient IoT

intrusion detection. Currently existing works deal separately with preprocessing, deep learning, blockchain, and federated learning. Recently proposed solutions such as Block Hunter[40] and Secure Intelligent Fuzzy Blockchain Framework[41] are based on the use of blockchain and federated learning techniques. The suggested framework unites SDN, DMMVI, CWO, DUSFAN, PCO, and SPBFT-LC in one IoT cybersecurity solution. The outcomes of an experiment validate that the approach offers robust and accurate attack detection that was easy for securing IoT systems against emerging cyber-attacks. Akhi et al.[42] presented a Residual-Temporal Convolutional Network (Res-TCN) that was designed to detect multiclass cyber-threats in a healthcare Internet of Things (H-IoT) environment. This approach employed SMOTE while training the algorithm to solve the issue of class imbalance and to test the Selective Forwarding, Man-in-the-Middle and DDoS attacks simultaneously within the H-IoT network topology. The accuracy achieved was 99.32%, which is a clear demonstration of the performance of temporal convolution for H-IoT intrusion detection. On the other hand, the current study aims at heterogeneous smart city IoT intrusion detection by using UNSW-NB15 and CIC-IDS2017 data sets alongside DMMVI, CWO, DUSFAN, PCO and SPBFT-LC algorithms.

Table 1 summarizes recent research on enhancing IoT cybersecurity in smart cities, presenting the authors, methods, advantages, and drawbacks. The studies utilize a variety of techniques, including ML (SVM, XGBoost), DL (DenseNet, LSTM-SVM), federated and incremental learning (FL, IL, and FDNN), unsupervised learning (OCSVM), and blockchain-based security (BFA). Unlike other IoT intrusion detection techniques currently available, the novel DUSFAN-PCO framework integrates DMMVI, CWO, DUSFAN, PCO, and SPBFT-LC to make up a comprehensive framework based on SDN. The integration of such components not only ensures the enhancement of each process but also highlights the importance and novelty of the suggested framework through an ablation study presented in "Statistical analysis and reproducibility" section. Such techniques help in the improvement of intrusion detection, proper attack prediction, data privacy protection, and provide scalability of distributed IoT networks. Hybrid and ensemble methods make them stronger and decrease false positives whereas federated and IL enables collaborative model updates in a secure manner. Nevertheless, the majority of approaches are characterized by the issues of high computation, complex implementation, the necessity to use large datasets, and inabilities to deal with changing or unbalanced attacks. In general, the table has identified the advantages and weaknesses of existing intelligent frameworks of IoT cybersecurity.

Problem statement and motivation

The fast proliferation of IoT devices in the Smart cities has resulted into complex and heterogeneous networks which are highly vulnerable to various cyber-attacks such as DoS, DDoS, APTs, and intrusion attacks, coupled with lack of assurance of data privacy and secure transactions. Current solutions, which include classical ML, DL, FL, and blockchain-based solutions, usually do not cope with real-time detection, scalability, adaptive threat processing, and unbalanced or noisy information. This paper will be inspired by these issues and suggest a Blockchain-Enhanced IoT Cybersecurity Framework in smart cities, where a Software-Defined Networking framework with a DUSFAN-PCO is used. The proposed framework enable a smart city IoT setting to have an intrusion detection system that is scalable, accurate, and trustworthy by incorporating DMO, CWO, DUSFAN, PCO, and a Blockchain-Based SPBFT-LC to manage the results securely.

Proposed methodology

The suggested methodology introduces the Blockchain-Enhanced IoT Cybersecurity Framework for Smart Cities, which incorporates Software-Defined Networking (SDN), deep learning, and optimization algorithms to secure and intelligently monitor smart cities. To begin with, the UNSW-NB15 and CIC-IDS2017 datasets are used to obtain the pre-processed data of the IoT network that undergoes reconstructing the missing information by using the Diffusion Model of Missing Value Imputation (DMMVI). The next step is the application of the

Table 1 Overview of the literature review.

Author	Method	Advantage	Drawbacks
More et al.[16]	SVM	Improved IDS performance; RM model effectively identifies cyber-attacks; Uses feature selection and correlation analysis	Limited to classical ML; May not handle evolving attack patterns effectively
Tareq et al.[17]	DenseNet	Detects multiple forms of cyber-attacks; Multi-class classification; Effective on large-scale experiments	Requires large datasets; High computational cost
Priya-darshini et al.[18]	FL	Preserves data privacy; Effective in distributed IoT; Scalable and resilient security	Training complexity; Communication overhead in federated setup
Mishra et al.[19]	LSTM-SVM	Accurate attack prediction; Combines time-series and classification methods; Blockchain ensures transaction authenticity	Pre-processing intensive; Requires blockchain infrastructure
Vikram et al.[20]	BFA	Strong data security via encryption and blockchain; Effective IIoT attack detection	High computational cost; Complex implementation
Mahdi et al.[21]	IL	Continuous learning improves detection; Secure data storage; Enhances IoT network-wide accuracy	Implementation complexity; Requires blockchain and distributed updates
Kaliyape-rumal et al.[22]	OCSVM	Detects unknown attacks; Handles unbalanced data; Scalable IDS	Unsupervised learning may produce false positives; Requires careful tuning
Awotunde et al.[23]	AI	High detection accuracy; Pre-processing improves feature extraction; Superior to existing methods	Model complexity; Requires careful integration of AI techniques
Saini et al.[24]	XGBoost	Efficient detection of APT attacks; Reduces false positives; Robust mechanism	Ensemble models are computationally intensive; May require large memory
Abbas et al.[25]	FDNN	Maintains data privacy; Accurate large-scale attack detection; Collaborative learning	Communication overhead; Requires distributed device coordination
Albanbay et al.[35]	CNN-based IDS	Preserves data privacy; Enables collaborative intrusion detection; Works well in distributed IoT systems	Communication overhead is higher; Performance is dependent on data distribution among clients
Huang et al.[36]	Hybrid DL with FL	Implements more accurate intrusion detection; Provides privacy; Centralized data sharing is minimized	Training process is more complex; Regular model synchronization is needed
Cao et al.[37]	Blockchain-enabled FIDS	Secures and makes data sharing tamper-proof; Increases trust and data privacy between IoT nodes	Involves overhead in terms of storage and consensus; Has scalability issues

Carpet Weaver Optimization (CWO) algorithm to conduct the process of feature selection to remove the most insignificant features and give more emphasis to those that render effective intrusion detection achievable.

The processed data is classified as either normal or malicious traffic by DUSFAN. The Pine Cone Optimization (PCO) improves the model weights for better performance in the detection process. Lastly, the SPBFT-LC blockchain platform is used to store the results of detection. The DUSFAN-PCO architecture is described in Fig. 1.

Research gap and novelty

Table 2 demonstrates the contradictory analysis to the earlier IDS methodologies which implement preprocessing, feature selection, deep learning, or blockchain security separately, this framework proposes an end-to-end methodology, whereby all the stages are tightly coupled in such a way that each stage is beneficial to the next stage, thus enhancing robustness and reducing false alerts.

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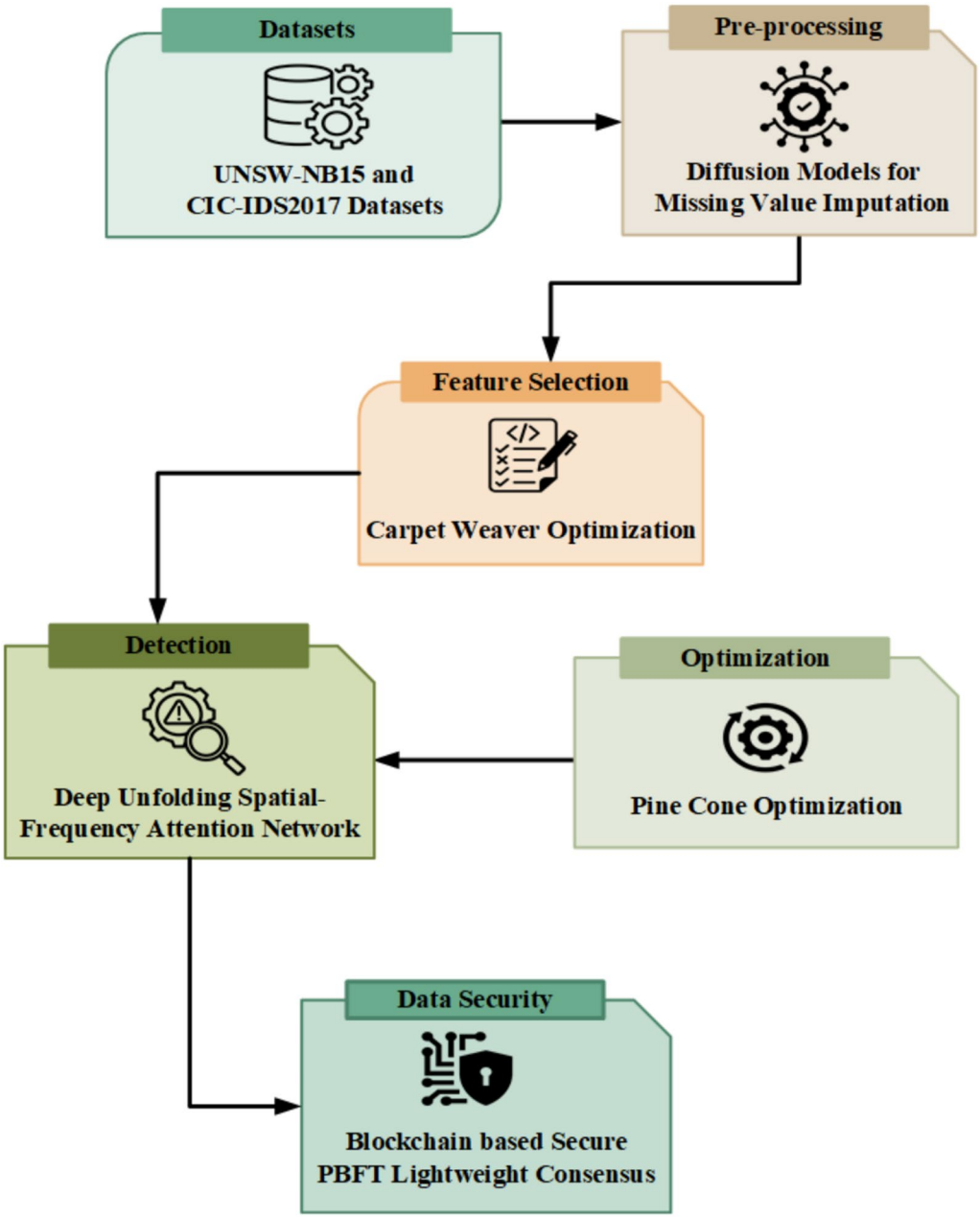


Fig. 1 Proposed methodology DUSFAN-PCO architecture

Table 2 Analysis of research gap and novelty.

Existing works	Limitation	Proposed solution
CNN/LSTM IDS	Ignore missing values	DMMVI reconstructs missing traffic
Feature selection	Static methods	CWO adaptive optimization
Attention networks	Spatial only	Spatial + Frequency Attention
Deep learning	Fixed weights	PCO adaptive optimization
Blockchain IDS	Heavy consensus	Lightweight SPBFT-LC

Data collection

To begin with, two benchmark datasets were used to collect network traffic data, namely UNSW-NB15 and CIC-IDS2017, to have comprehensive and realistic testing of intrusion detection. The UNSW-NB15 data set includes both normal and malicious traffic, featuring modern types of attacks (Fuzzers, Exploits, DoS, and Backdoors), whereas the CIC-IDS2017 data set contains real network flows that include attacks such as DDoS, port scan, and Web attacks. The preprocessing step then takes the received data and filters it to remove noise, replace missing data and normalize the feature distributions to fit the model.

Preprocessing using diffusion models for missing value imputation

In this part, DMMVI[26] is applied to increase the quality of the data of IoT network data through filling the missing and noisy values. This technique reconstructs missing data via a diffusion algorithm, eliminates the duplicates, and normalizes the features. Preprocessing ensures the creation of full and consistent data sets as well as the preservation of traffic patterns. The diffusion model is mathematically represented as shown in Eq. (1).

$$q\phi\left(y_{u-1}^{ub}\left|y_{u}^{ub},y_{0}^{dp}\right.\right)=M\left(y_{u-1}^{ub};\beta_{\phi}\left(y_{u}^{ub},u\left|y_{0}^{dp}\right.\right),\rho\,\mathrm{E}\right) \tag{1}$$

where, y_{u-1}^{ub} denotes reconstructed data sample at the previous time step, y_{u}^{ub} denotes current noisy, y_{0}^{dp} denotes clean data distribution prior to diffusion, β_{ϕ} is the learned denoising coefficient, $\rho\,\mathrm{E}$ denotes noise term that ensures stability during the reverse diffusion, M denotes mean function governing the Gaussian transition process between diffusion steps. Lastly, DNNVI is employed to guarantee that the full traffic data of the IoT networks is properly reassembled, denoised and normalized to give a clean and consistent data which is much more reliable and one that works better at SDN-based intrusion detection framework implementation in cybersecurity application in smart cities. The normalized and reconstructed data is then given to the feature selection step.

Feature selection using carpet weaver optimization

In this section, Carpet Weaver Optimization (CWO)[27] algorithm is adopted to obtain the significant features for the preprocessing of IoT network data set. The motivation behind using this algorithm is that the carpet weaving technique involves the combination of several features and the choice of the best combination that can be obtained by evaluating different feature combinations. This method involves an iterative updating of feature subsets on the basis of their fitness scores. CWO is capable of reducing the computation cost and increasing the accuracy of intrusion detection.

Step 1: Initialization.

The population of candidate solutions is created randomly in the search space defined in the first step, which is the initialization step. Every individual member of the population is a possible solution of the optimization problem. Such randomness is needed to achieve diversity and make the search in the solution space productive. It is provided as Eq. (2).

$$Y=\begin{bmatrix} y_{1,1} & \cdots & y_{1,D} & \cdots & y_{1,M} \\ \vdots & \ddots & \vdots & \iddots & \vdots \\ y_{j,1} & \cdots & y_{j,D} & \cdots & y_{j,M} \\ \vdots & \iddots & \vdots & \ddots & \vdots \\ y_{n,1} & \cdots & y_{n,D} & \cdots & y_{n,M} \end{bmatrix}_{n\times M} \tag{2}$$

where,n indicates the quantity of potential solutions, m is number of decision variables, $y_{j,M}$ represents the value of the jth variable in the candidate solution, Y denotes candidate solution in the population.

Step 2: Random generation.

The initial population is used to generate a random solution that is used to start the iterative search process. This helps introduce variability and avoids early convergence to suboptimal solutions.

Step 3: Fitness function.

It generates an initial value solution in a random manner. It is represented as in Eq. (3),

$$Fitness\,Function = Optimize\,(Selecting\,relevant\,features) \tag{3}$$

Step 4: Exploration phase.

The exploration stage emulates the wide-ranging weaving movement to cover the entire feature space the world over. The candidate solutions are transferred to new positions when they enhance the objective function, and make sure that promising regions are covered. The update of positions of candidate feature subsets, as in Eqs. (4) and(5), is performed stochastically to find globally optimal combinations of features.

$$y_{j,k}^{Q1} = y_{j,k} + (1-2s).\left(y_{q,k} - J.y_{j,k}\right) \tag{4}$$

$$Y_j = \begin{cases} Y_j^{Q1}, G_j^{Q1} \le G_j \\ Y_j,\ else \end{cases} \tag{5}$$

where, J denotes randomly selected integer, either 1 or 2, controlling the scaling effect, s denotes random number in the interval [0, 1], introducing stochasticity, G_j indicates the value of the objective function at the present point y_j,G_j^{Q1} denotes objective function value at the new position $y_{j,k}^{Q1}$,Y_j^{Q1} denotes new position of the jth member in the first phase of CWO, $y_{j,k}$ denotes the jth dimension of the weaving pattern.

Step 5: Exploitation phase.

The algorithm makes small refined changes on the solution to improve the quality of solutions during exploitation. Better feature subsets are introduced based on the old ones and the selection is refined to yield better intrusion detection. It is represented in Eq. (6) and (7).

$$y_{j,k}^{Q2} = \left(1 + \frac{(1-2s)}{u}\right).y_{j,k} \tag{6}$$

$$Y_j = \begin{cases} Y_j^{Q2}, G_j^{Q2} \le G_j \\ Y_j,\ else \end{cases} \tag{7}$$

where, $y_{j,k}^{Q2}$ denotes new calculated position of the jth CWO member in the kth dimension. It optimises the positions locally to move towards the optimal set of features, making sure that the accuracy is high and the redundancy of features is minimal.

Step 6: Termination.

The iteration is continued until the convergence criteria are satisfied, e.g. the maximum number of iterations is reached, or the objective function has a small improvement. The last subset is the most informative to the data set of the IoT traffic. Lastly, CWO finds the best combination of features that maintain important security properties and increases the generalization ability of the DL model. The features chosen are then passed to the detection step.

Detection using deep unfolding spatial-frequency attention network

In this paper, Deep Unfolding Spatial-Frequency Attention Network (DUSFAN)[28,29] has been employed for classifying IoT network traffic into two categories; one being normal and other being attacks such as DDoS, DoS, PortScan, and Fuzzers. DUSFAN employs Gates Controlled Deep Unfolding Network (GCDUN) along with Spatial-Frequency Convolutional Self-Attention Network (SFCSAN) to extract features both from space and frequency. Deep unfolding helps to make better predictions, whereas gates eliminate unnecessary features.

Self-attention helps in highlighting important traffic features that help to increase the accuracy of classification. Entropy value defined in Eq. (8) defines the amount of uncertainty in network traffic, with higher entropy values showing abnormal traffic and lower entropy values showing normal traffic.

$$EF = -\int_{-\infty}^{+\infty} g(y)\log(g(y))\,ey \tag{8}$$

where, $g(y)$ denotes probability distribution of traffic features. Equation (9) makes sure that the weights of the network are updated to reconstruct with minimum error, and to maintain smoothness that is very important in detecting in dynamic IoT environment.

$$w^{(l)} = \arg\min_{w} \frac{1}{2}\left\|w - s^{(l)}\right\|_2^2 + \mu G(w) \tag{9}$$

where, $w^{(l)}$ denotes the optimized weight at layer ,$s^{(l)}$ denotes target feature at the same layer, μ denotes regularization coefficient, $G(w)$ denotes regularization function ensuring stability. The coefficient measures the importance of every feature in comparison to the other features and therefore the network is able to concentrate on the most significant portions of the traffic in order to classify it. It is as shown in Eq. (10).

$$\alpha_{k,j}^{c} = \frac{\exp\left(g\left(V_j^c\right)^T h\left(V_k^c\right)\right)}{\sum_j \exp\left(g\left(V_j^c\right)^T h\left(V_k^c\right)\right)} \tag{10}$$

where, V_j^c and V_k^c is denotes feature vectors from spatial-frequency, $g\left(V_j^c\right)^T$ denotes learnable transformations, $\alpha_{k,j}^{c}$ denotes the attention coefficient. Equation (11) is the summation of weighted contribution of the relevant features, which results in a complete representation of local and global traffic patterns.

$$p_k^c = w_c\left(\sum_{j=1}^{e_y^c} \alpha_{k,j}^{c}\, i\left(V_j^c\right)\right) \tag{11}$$

where, $i\left(V_j^c\right)$ denotes input feature mapping function, p_k^c denotes aggregated feature representation for the k th feature class in c. Equation (12) update enables the network to improve predictions layer by layer and in the process enhance the identification of tiny anomalies as it progresses.

$$w^{(l)}, i^{(l)} = I_{gcim}\left(w^{(l-1)}, i^{(l-1)}, w\right) \tag{12}$$

where, $w^{(l-1)}$ and $i^{(l-1)}$ denotes previous layer's weights and transformations, I_{gcim} denotes gated unfolding update function. The resulting enhanced feature representation according to Eq. (13) is a combination of attention-weighted and original features to obtain the best classification accuracy.

$$z_j^c = \lambda^c\, p_j^c + V_j^c \tag{13}$$

where, λ^c denotes scaling factor controlling the contribution of the attention mechanism. The combination of DUSFAN and GCDUN and SFCSAN is necessary, so that the IoT traffic may be analyzed in spatial and frequency domain, attention mechanisms may emphasize important characteristics, and deep unfolding may enhance the detection progressively. The framework can identify various types of attacks in real-time by integrating these mechanisms to ensure structural and temporal relationships that ensure SDN-based cybersecurity in smart cities. Lastly, DUSFAN produces correct, multi-class traffic forecasts, which are dependable to feed the next optimization stage.

Optimization using pine cone optimization

In this section, the PCO[30] is employed to optimize the weight parameters λ and μ of DUSFAN. Mimicking the natural process of pine cone seed dispersal and germination, PCO mimics how seeds are dispersed and grow preferentially in favorable locations, thus optimizing resource utilization.

Step 1: Initialization.

At the initialization step, a population of candidate solutions is randomly drawn, which embodies various possible configurations for DUSFAN weight parameters. Each candidate is similar to a pine seed scattered in various locations, guaranteeing a broad coverage of the search space.

Step 2: Random generation.

The generation of randomness is done through creating new solution candidates through making slight changes to the initial weights. This is a representation of the natural distribution of pine seeds to different places in order to look for better conditions for growing. The generation of this diversity ensures that the algorithm does not get trapped in the local optima.

Step 3: Fitness function.

The objective function is used to determine the fitness function. Equation (14) provides an expression for it.

$$Fitness function = [optimizing\ (\lambda\ and\ \ \mu)] \tag{14}$$

Step 4: Pine cone dispersal by gravity optimizing (λ).

In this step, pine cones are moved in the search space, simulating the effect of gravity to enhance local exploration. The dispersal adjusts positions toward promising areas while maintaining randomness to avoid premature convergence. This approach balances exploration and exploitation by guiding solutions toward optimal regions based on their relative fitness. It is expressed in Eq. (15).

$$DY_{k,j}^{new} = \begin{cases} VY_{\ j} + \lambda x_1 \times S_1 \times \left(S_2 \times \left(ub_t^j - lb_t^j - VY_{\ j}\right)\right), & if\ \ Control_{parameter=0} \\ DY_{\ k,j} + x_1 \times S_1 \times \left(S_2 \times \left(ub_t^j - lb_t^j - V_{pop_{all,s1}}\right) - Vbest_{y,j}\right), & otherwise \end{cases} \tag{15}$$

where, ub_t^j indicates the super-cube's upper bound, S_1 and S_2 normal random numbers between 0 and 1 lb_t^j indicates the super-cube's lower bound, $DY_{k,j}^{new}$ is the new position of the pine cone, $V_{pop_{all,s1}}$ denotes randomly selected solution, $DY_{\ k,j}$ indicates the pine cones most recent location, $VY_{\ j}$ denotes position of the tree.

Step 5: Pollination optimizing (μ).

In this step, the pollination process imitates the transfer of agents from male cones to female cones, influenced by natural wind flow. It makes sure that it is explored whereby agents are able to move within the same solution space or towards others. This mechanism increases diversity and possibility of coming up with the best solutions. It is expressed in Eq. (16).

$$DY_{k,j}^{new} = DY_{\ k,j} + 0.5 \times \phi_{s_1} \times \mu\left(Vbest_k - DY_{\ S_1,j}\right) + 0.5 \times \phi_{s_3} \times \left(Vbest_k - DV_{s_{3,j}}\right) \tag{16}$$

where, ϕ_{s_1} represents likelihood of pine cones being successfully pollinated, $Vbest_k$ is current best position.

Step 6: Termination.

The PCO step 3 is reiterated until the weight parameters that are used in the DUSFAN reach the optimal solution. This will make the model have maximum detection accuracy and robustness of various classes of IoT network traffic. The streamlined outcomes are then sent to the blockchain stage to store, authenticate and handle the IoT network information in the SDN-enabled smart city network. The overall process flow of the PCO is illustrated in Fig. 2.

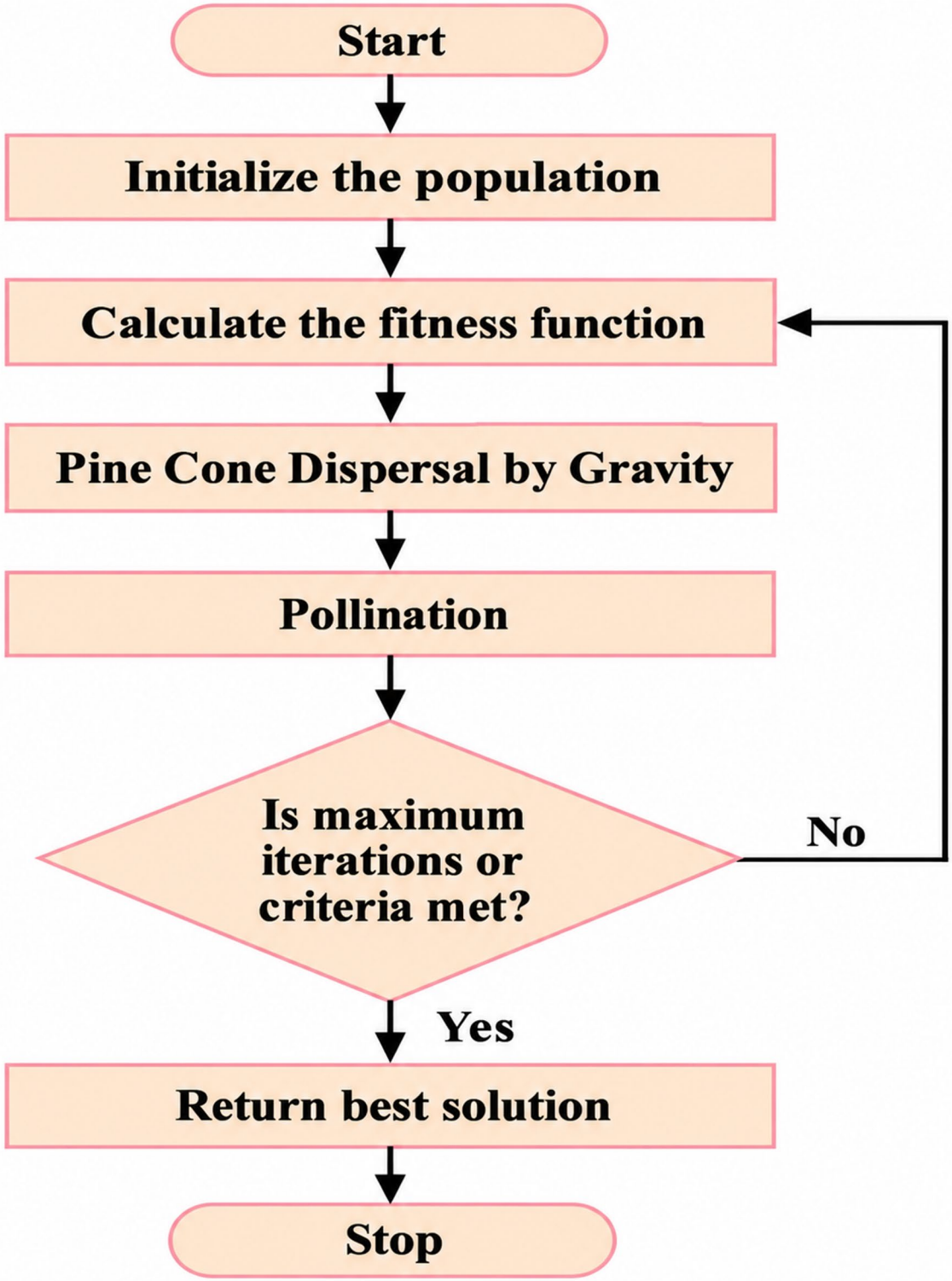


Fig. 2 Flowchart of PCO

Data security using blockchain-based secure PBFT lightweight consensus

The IoT network traffic outputs and optimized DUSFAN outputs are then secured by the Blockchain-Based Secure PBFT Lightweight Consensus (SPBFT-LC)[31] to ensure the integrity, transparency, and security of data against unauthorized use, including protection against data integrity, data transparency, and unauthorized access, with the results of the classification. The SPBFT-LC design creates a decentralized registry in which all authenticated events within the IoT network are permanently stored, providing authentication, confidentiality, and immunity to interference from adversaries. In the consensus protocol, the central node is selected using the formula described in Eq. (17).

$$\mathrm{Pr}\,imary_node = w_num \mod |D| \tag{17}$$

where, w_num denotes current sequence number, D denotes total number of participating nodes in the network. Such a choice guarantees the equity and determinism of leadership rotation so that one node does not dictate block offers. Equation (18) is used to measure the reliability and contribution of each node by using a utility score.

$$U_{qr} = \frac{\max\left(T_{qr}, 0\right)}{\sum_r \max\left(T_{qr}, 0\right)} \tag{18}$$

where, T_{qr} denotes the number of successful transactions validated by node, U_{qr} denotes quantifies the trustworthiness and performance of each node, helping the consensus process prioritize reliable participants. The utility scores are summed up to obtain network-wide reliability, as represented by the interactions between all the nodes, as in Eq. (19).

$$U_{qr} = \sum_s U_{qs} U_{sr} \tag{19}$$

Where, U_{qs} denotes utility of node s denotes evaluated by node q. This aggregation must ensure that there is consensus-based decision-making, that is, the power of collective trust, rather than the behavior of isolated nodes, thereby improving robustness to faulty or malicious nodes. With a mixture of primary node selection, utility-based scoring, and interaction aggregation, SPBFT-LC is a lightweight and secure consensus mechanism that guarantees integrity, authenticity, and confidentiality of records related to the traffic of the IoT network. This becomes particularly significant in smart cities where the SDN allows making data available and reliable in terms of the management of network resources and prevention of cyber threats. Lastly, the Blockchain-Based Secure PBFT Lightweight Consensus will safeguard all perceived and optimized events within the IoT network, ensuring that these events are securely stored in an immutable ledger and provide a reliable basis for smart city cybersecurity.

Computational complexity analysis

The computational complexity of the proposed approach was calculated to find out whether it is capable of functioning in a real-time mode during intrusion detection in smart city IoT environments. The complexity of each module is given in Table 3.

Table 3 Computational complexity.

Module	Complexity
DMMVI	O(nT)
CWO	O(N × D × I)
DUSFAN	$O(L \times F^2)$
PCO	O(P × I × W)
SPBFT-LC	$O(n^2)$

From the observed execution time, the computational power to achieve near-real-time computation is seen in the present experimental setup. Nevertheless, since the current assessment employs benchmark datasets rather than actual Internet of Things (IoT) traffic, the current latency observed cannot be considered as proof of real-time deployment. Run time performance of each module is given in Table 4. The proposed framework takes around 114 ms to perform pre-processing, feature selection, intrusion detection, optimization, and blockchain validation and thus can be used in real-time IoT security application. It can be observed from the above result that the proposed framework has moderate complexity with low run time. Thus, the proposed framework can be used in near real-time IoT security applications for smart cities.

Results and discussion

The suggested DUSFAN-PCO model is found to be better than others in detecting diverse IoT cyberattacks with good accuracy, precision, and robustness. Testing with UNSW-NB15 and CIC-IDS2017 datasets finds that it reports high detection and low false positives compared to existing methods. PCO fine-tuning and CWO feature selection enhance the learning and generalization abilities of the model. In addition, the SPBFT-LC mechanism on a blockchain enables the secure and tamper-evident storage of detection results, which proves the effectiveness of the framework in real-life scenarios within the ecosystem of smart cities IoT.

Experimental setup

The Keras deep-learning library and Python 3.8 were employed to execute the introduced DUSFAN-PCO algorithm. In order to speed up the training and testing processes, the tests were carried out on a computer with an Intel Core i7-7700 processor and a graphics card. The framework accuracy, stability, and computing efficiency were critically tested within the IoT cybersecurity scenario in the training and testing processes of the framework with benchmark datasets, UNSW-NB15, and CIC-IDS2017. Cross-domain generalization was assessed by training the model on [UNSW-NB15/CIC-IDS2017] and testing it without retraining on [external dataset]. The method of train-on-one and test-on-another allows one to check how the learned representation generalizes outside the source domain distribution. These simulation parameters are specified in Table 5.

Experimental and hyperparameter settings

Figure 3 depicts the work process of the introduced DUSFAN-PCO architecture for IoT intrusion detection. Firstly, data regarding the IoT network traffic are gathered and preprocessed by implementing data cleansing, feature extraction, and normalization. Afterwards, the preprocessed data are provided to the DUSFAN architecture where the spatial frequency attention mechanism and deep unfolding are applied to extract traffic features. Pine Cone Optimization (PCO) algorithm is employed to optimize the network parameters in order to enhance the detection performance. The classification is performed by the Softmax layer to categorize traffic as either normal or an attack.

Table 4 Runtime table.

Module	Time (ms)
Preprocessing	18
Feature Selection	27
Detection	42
Optimization	16
Blockchain	11

Table 5 Experimental parameters and software configuration.

Parameters	Description	Actual Software URL
Software	Python 3.8	https://www.python.org/downloads/release/python-380/
Deep Learning Framework/Library	TensorFlow 2.12 / Keras 2.12	https://pypi.org/project/tensorflow/2.12.1/https://pypi.org/project/keras/2.12.0/
Operating System	Windows 10	https://www.microsoft.com/software-download/windows10
Processor	Intel Core i7-7700	https://www.intel.com/content/www/us/en/products/sku/97128/intel-core-i77700-processor-8m-cache-up-to-4-20-ghz/specifications.html
Graphics Processing Unit	GPU-based experimental configuration	https://developer.nvidia.com/cuda-toolkit
Data Type	IoT Network Traffic Data	https://research.unsw.edu.au/projects/unsw-nb15-dataset
Datasets	UNSW-NB15 and CIC-IDS2017	https://research.unsw.edu.au/projects/unsw-nb15-dataset https://www.unb.ca/cic/datasets/ids-2017.html
Data Partitioning	80% Training / 20% Testing	https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html
Missing-Value Handling	Deep Missing-Value Imputation (DMMVI)	https://scikit-learn.org/stable/modules/impute.html
Feature Optimization	Carpet Weaver Optimization (CWO)	No official URL; proposed algorithm
Proposed Neural Network	Deep Unfolding Spatial-Frequency Attention Network (DUSFAN)	https://www.tensorflow.org/api_docs
Parameter Optimization	Pine Cone Optimization (PCO)	No official URL; proposed algorithm
Network Architecture	Software-Defined Networking (SDN)	https://mininet.org/
Data Security and Storage Layer	Blockchain-based SPBFT-LC	https://hyperledger-fabric.readthedocs.io/
Class Imbalance Handling	SMOTE, minority oversampling, class weighting, weighted cross-entropy, and balanced batch construction	https://imbalanced-learn.org/stable/
Statistical Analysis	Wilcoxon signed-rank test, McNemar's test, Cohen's d, and confidence-interval analysis	https://docs.scipy.org/doc/scipy/reference/stats.html https://www.statsmodels.org/
Performance Metrics	Accuracy, precision, recall, F1-score, computational complexity, and runtime	https://scikit-learn.org/stable/modules/model_evaluation.html

Training was conducted on the proposed DUSFAN-PCO with the set of hyperparameters as provided in Table 6. The optimization process was performed via the Adam optimizer with a learning rate of 0.0001. Training involved 100 epochs and a batch size of 64.

The activation function used in the hidden layers was ReLU while that used in the output layer for multi-class classification was Softmax. The Categorical Cross-Entropy was the chosen loss function for the training process.

Dataset description

This portion offers a comprehensive summary of datasets utilized in the research is described below.

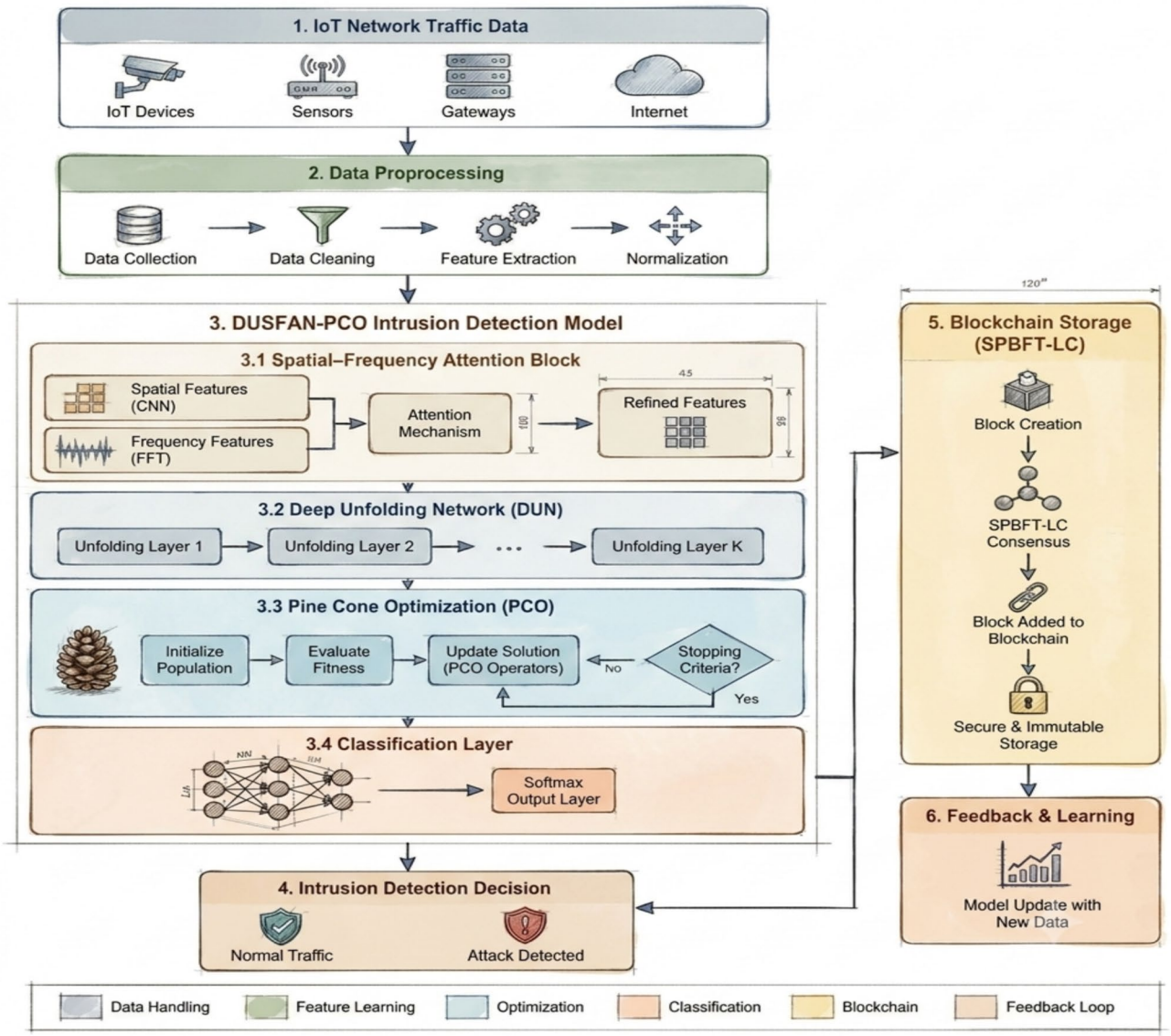


Fig. 3 Overall architecture of the proposed model along with experimental setup

Table 6 Set of hyperparameters.

Parameter	Value
Epochs	100
Batch Size	64
Optimizer	Adam
Learning Rate	0.0001
Dropout	0.3
Activation	ReLU
Loss	Categorical Cross Entropy
Train/Test	80/20

Table 7 UNSW-NB15 Dataset Splitting.

Classes	Total No. of instances	Training (80%)	Testing (20%)
Normal	93,000	74,400	18,600
Backdoor	2329	1863	466
Analysis	2677	2142	535
Fuzzers	24,246	19,397	4849
Shellcode	1511	1209	302
Reconnaissance	13,987	11,190	2797
Exploits	44,525	35,620	8905
DoS	16,353	13,082	3271
Worms	174	139	35
Generic	58,871	47,097	11,774
Total	257,673	205,139	52,534

Table 8 CIC-IDS2017 dataset splitting.

Classes	Total No. of instances	Training (80%)	Testing (20%)
Benign	2,273,097	1,818,477	454,620
DDOS	128,027	102,421	25,606
DOS	252,661	202,128	50,533
Bot	1966	1572	394
Patator	13,835	11,068	2767
Heartbleed	11	8	3
Infiltration	36	28	8
PortScan	158,930	127,144	31,786
Web Attack	2180	1744	436
Total	2,830,743	2,263,590	567,153

Dataset 1: UNSW-NB15 dataset

The UNSW-NB15 data set[38] is widely employed in testing intrusion detection models since it contains a well-balanced mixture of normal and attacking traffic, which was induced in a controlled environment. It encompasses a variety of attacks such as DoS, Exploits, Fuzzers, and Generic; hence, it offers end-to-end coverage of the modern-day cybersecurity attacks. Table 7 shows the distribution of 257,673 total samples to classes, with 80 percent used for training and 20 percent used for testing. The provided partitioning provides an adequate number of data for all classes to ensure that the proposed DUSFAN-PCO framework learns various attack patterns and acquires correct and accurate detection performance.

Dataset 2: CIC-IDS2017 dataset

The CIC-IDS2017 dataset[39] is a rich benchmark for intrusion detection research, featuring realistic network traffic mimicking recent attack patterns and typical user behavior. It supports multiple attack types, such as DDoS, DoS, Botnet, Port Scan, and Web attacks, making it possible to utilize in testing multi-class classification models. Table 8 highlights how the dataset is partitioned into two sets with a total of 2,830,743 instances, out of which 80% and 20% are used in the training and testing sets, respectively. This high-quality partition guarantees that the designed DUSFAN-PCO framework will be able to identify intricate attack patterns and distinguish accurately between benign and malicious IoT network traffic.

Table 9 Performance metrics.

Metrics	Formulation	Metrics	Formulation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Precision	$\frac{TP}{TP+FP}$
Sensitivity	$\frac{TP}{TP+FN}$	Specificity	$\frac{TN}{TN+FP}$
ROC	$TPR = \frac{TP}{TP+FN} FPR = \frac{FP}{FP+TN}$	F1-score	$\frac{2\times Precision\times Sensitivity}{Precision+Sensitivity}$
Blockchain Secure Storage Metrics			
Latency	$Current\ completion\ time - Expected\ completion\ time$	Throughput (Tp)	$Tp = \frac{Total\ Transactions}{Total\ time}$

Table 10 Memory, CPU, and blockchain security metrics of the optimized DUSFAN-PCO model.

Number of nodes	Memory usage (%)	CPU usage (%)	Security integrity (%)
50	8.39	11.23	99.25
100	13.46	13.64	99.30
200	14.81	17.71	99.51
500	21.37	23.63	99.60
1000	29.94	32.19	99.92

Performance metrics

The proposed model DUSFAN-PCO is tested using Specificity, Accuracy, Sensitivity, F1-score, Precision, and ROC measures, which determine the capacity of the model to detect attacks in the IoT network. These performance measures are shown in Table 9.

True Positive (TP): This is a proportion of the amount of attack instances the model classifies as attacks.

True Negative (TN): TN is the count of instances that are normal and the actual instances that are recognized by the model as being normal.

False Positive (FP): The false positive is the count of normal cases that are classified as attacks.

False Negative (FN): This is the amount of attack instances that are wrongly identified as normal.

Performance analysis of the blockchain-enhanced proposed DUSFAN-PCO model

The performance of the suggested DUSFAN-PCO approach is evaluated on Dataset 1 and Dataset 2 to assess its effectiveness in IoT intrusion detection for smart city environments. In Dataset 1, the framework is compared to baseline models such as SVM[16], DenseNet[17], FL[18], LSTM-SVM[19], and BFA[20], whereas in Dataset 2; it will be compared to IL[21], OCSVM[22], AI[23], XGBoost[24] and FDNN[25]. Findings indicate that DUSFAN-PCO is superior to such approaches in F1-score, precision, sensitivity, accuracy, specificity, and ROC, making an effective distinction between normal and attack traffic. Also, the secure storage using blockchain is assessed by PoA[32], P2P[33], and PoS[34], with low latency and high throughput as the results of the detection are securely managed. Below, the results of corresponding performance and visual representations are shown in Table 10.

Table 6 indicates the memory, CPU, and blockchain security measures of the optimized model of DUSFAN-PCO with varying levels of IoT nodes. With the number of nodes, memory and CPU consumption also increase gradually, whereas security integrity is always high (99.25%-99.92%). This shows that the model is effective in scaling up to a larger network and offers secure and reliable blockchain storage.

Performance analysis using dataset-1

The purpose of evaluating the proposed DUSFAN-PCO framework on Dataset-1 is to determine its ability to identify and categorize IoT network threats in smart city settings. The associated findings and performance results are summarized below:

Figure 4 below presents the confusion matrix of the Dataset-1 on the DUSFAN-PCO framework. The actual classes are enclosed in the rows, and the expected classes are enclosed in the columns. The elevated values in

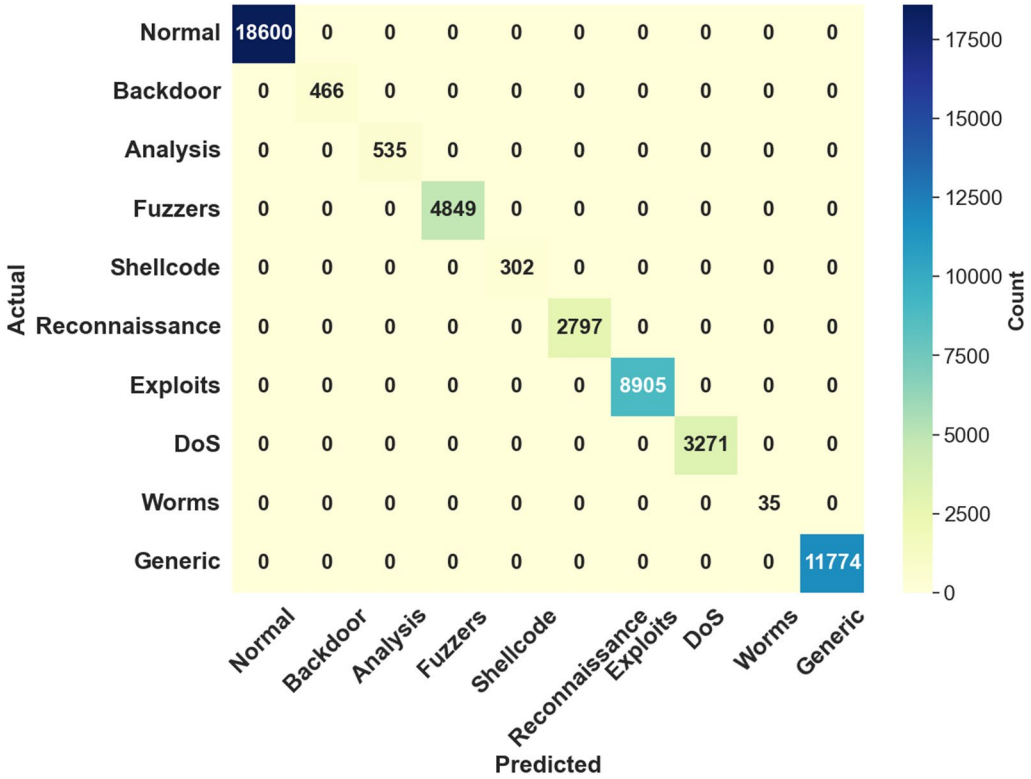


Fig. 4 Confusion matrix of DUSFAN-PCO on dataset-1

the diagonal factor imply that the model is quite accurate in recognizing normal traffic and the types of attacks, including Exploits, Fuzzers, and Generic attacks, among others, with a few misclassifications. This proves the effectiveness of this framework in identifying and separating various threats of the internet of things network.

Figure 5 shows the proportion of testing cases in various classes of attacks in Dataset 1. It is evident that the normal class has the largest number, then Generic, Exploits, and Fuzzers. By comparison, the Backdoor, Analysis and Shellcode classes contain fewer samples. Such an imbalance corresponds to the heterogeneous and uneven nature of the traffic of IoT network, which poses difficulties in the precise detection of intrusion and highlights the need to use efficient models such as DUSFAN-PCO to deliver balanced and dependable detection results across all types of attacks.

Figure 6 shows the number of tests of various types of attacks in Dataset-1. The frequency histogram indicates that the majority of the types of attacks fall within the lower end of the testing counts and it is more so in the range of 0–5,000 and 0–2,500. This indicates a significant class imbalance, where several attack categories have far fewer testing samples compared to others, which can impact both the training and evaluation of the intrusion detection technique.

Figure 7 shows the training dynamics of the proposed DUSFAN-PCO model after 100 epochs on Dataset-1. Figure 6a depicts the Accuracy curve, it is observed that the training as well as the validation accuracy rises sharply and reaches a high point (near 1.0 or 100%), after approximately 50 epochs, which is a high learning performance. Concurrently, Fig. 6b, the Loss curve, demonstrates a sharp decrease in training and validation loss, dropping close to 0 after approximately 50 epochs, signifying that the model has effectively minimized its error. The generally similar trends between the training and validation curves, with only minor volatility in the validation loss, suggest that the model is learning well without severe overfitting.

Figure 8 presents the ROC curves for the proposed DUSFAN-PCO technique's multi-class classification on Dataset-1. The figure distinguishes among the Normal traffic and 9 different types of attacks: Analysis, Worms,

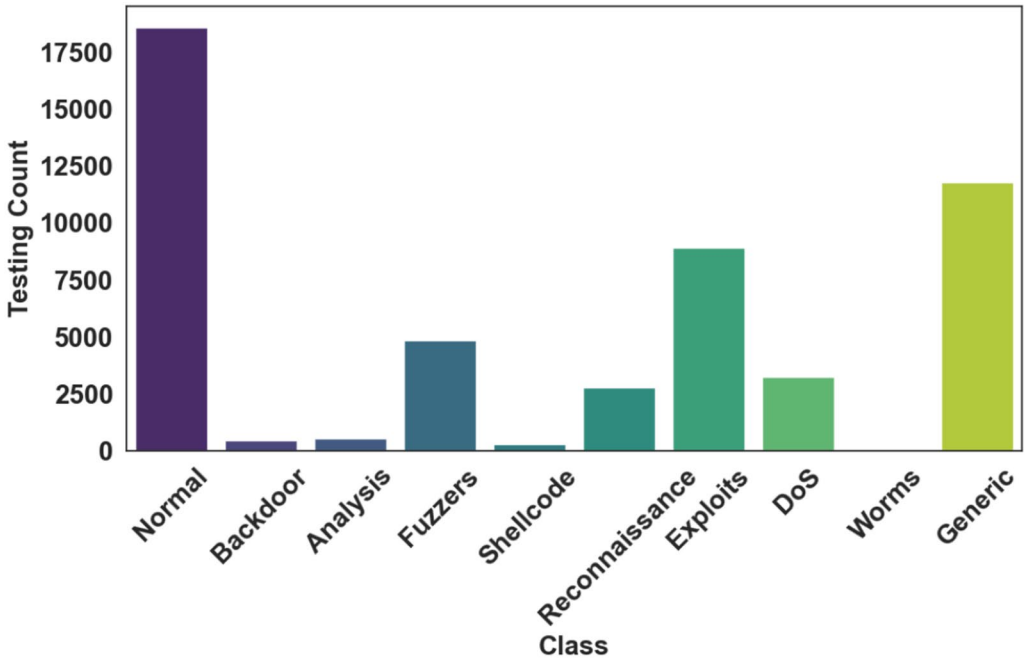


Fig. 5 Class-wise distribution of testing samples in dataset 1

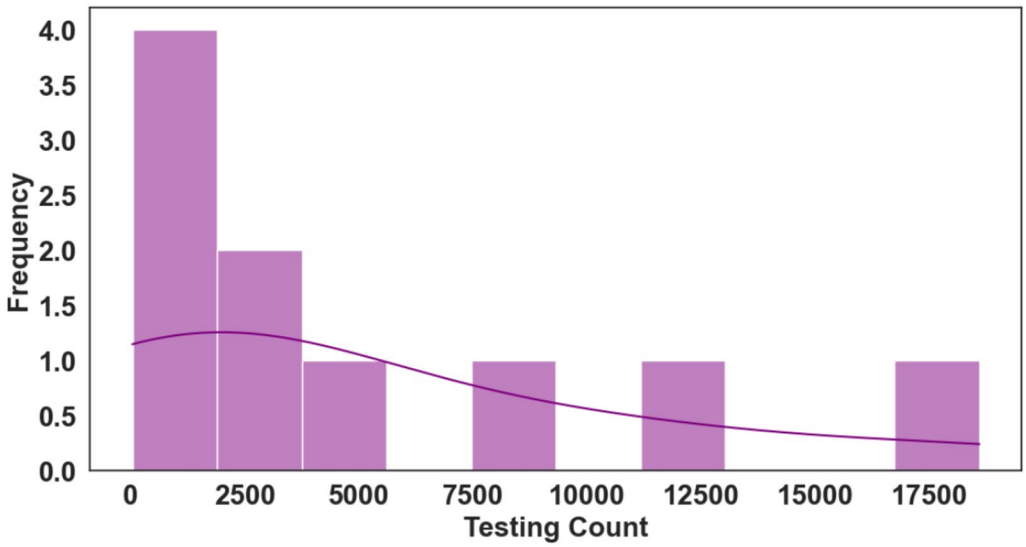


Fig. 6 Distribution of testing counts for dataset-1

Shellcode, Backdoor, Fuzzers, Exploits, DoS, Reconnaissance and Generic. The curves are curves of True Positive Rate (Sensitivity) vs. False Positive Rate at different threshold settings. The Area under the Curve (AUC) values are extremely high i.e. 1.00 in most of the classes and 0.99 in the others i.e. almost perfect discriminative capability. These findings confirm the excellent efficacy of the technique in precise identification and categorization of various attacks of the IoT network in smart city settings.

Figure 9 compares the proposed DUSFAN-PCO framework to the five existing intrusion detection models, i.e., SVM, DenseNet, FL, LSTM-SVM, and BFA on Dataset-1. The bar chart will compare the main performance indicators, such as, Recall, Precision, Accuracy, and F1-score.e. The findings are a clear indication of superiority

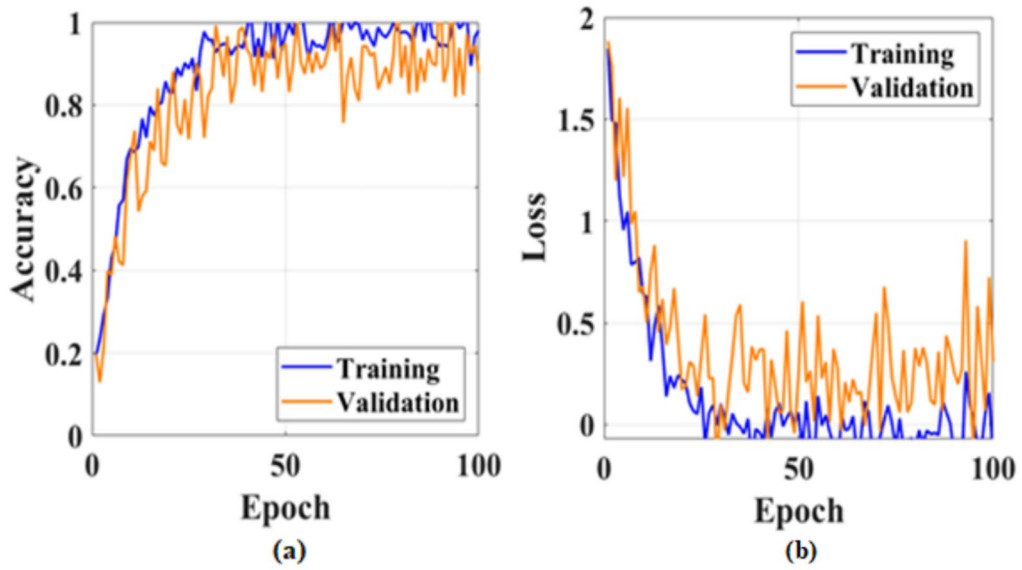


Fig. 7 Proposed (**a**) Accuracy (**b**) Loss curve on dataset 1

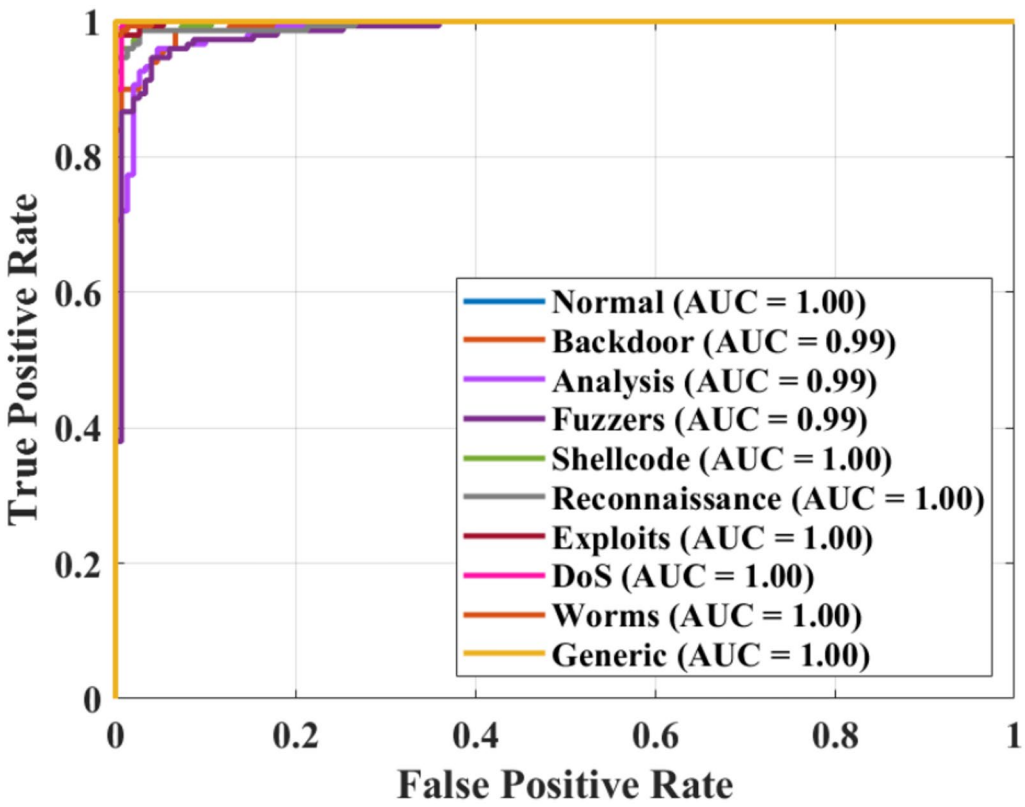


Fig. 8 Multi-class ROC curve with AUC on dataset-1

of the proposed framework that obtains performance values of close to 100 percent (99.6 and above) in all the metrics. Conversely, the lowest performing models have much lower scores, and their best scores are usually between 91 and 96. The results of the study indicate that the performance of the DUSFAN-PCO model in detecting IoT intrusions within a smart city network has been significantly improved and the model has better detection

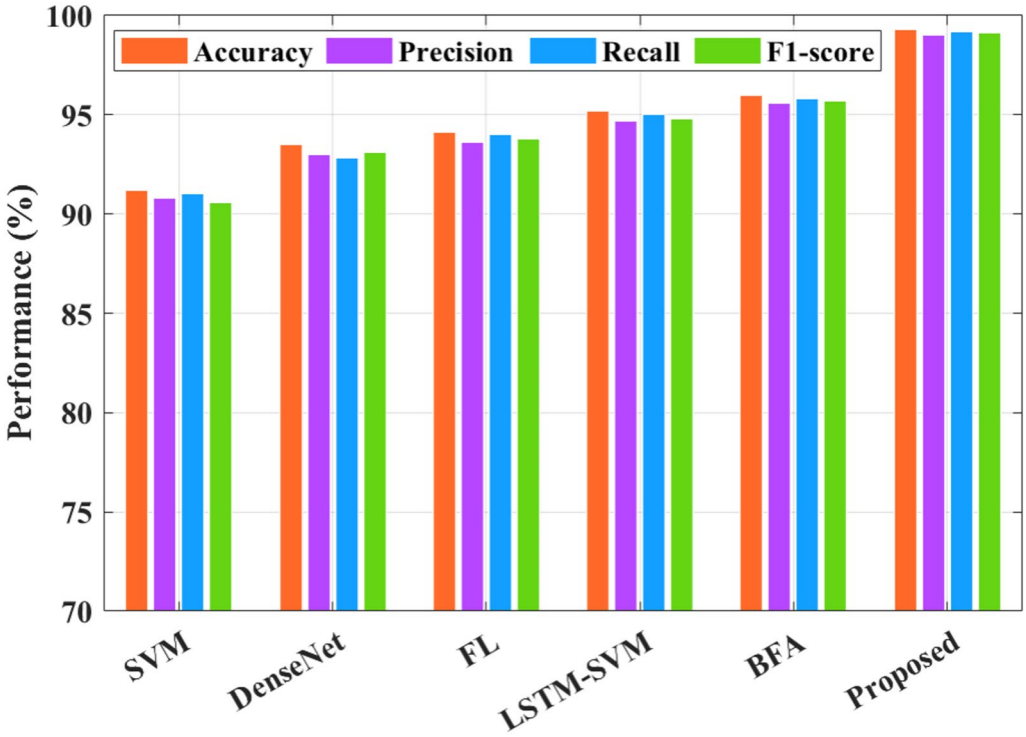


Fig. 9 Comparative performance analysis of DUSFAN-PCO with existing models on dataset-1

Table 11 Comparison of blockchain-based secure storage mechanisms for IoT.

Method	Throughput	Latency	Storage/Block (KB)	Consensus time (ms)
PoA[32]	42	110	46	52
P2P[33]	75	100	39	44
PoS[34]	89	90	34	36
Proposed	**99**	**70**	**28**	**24**

capabilities. The scalability testing of the blockchain was done by changing the number of participating nodes from 50 to 1000. All these parameters of CPU usage, memory usage, throughput, latency, storage overhead, consensus duration, and security integrity were studied using different node numbers. The workload used in all tests was kept the same along with the SPBFT-LC consensus algorithm.

The performance analysis of various Blockchain-based secure storage techniques in the IoT environment through Dataset 1 has been shown in Table 11. In comparison to PoA, P2P, and PoS approaches, the performance of the proposed technique SPBFT-LC with DUSFAN-PCO provides better performance in terms of maximum throughput (99 TPS) and minimum latency (70 ms). In addition, the lowest storage overhead (28 KB/block) and consensus time (24 ms) have been attained by the proposed technique. This shows that there is efficient communication and faster block processing.

Performance analysis using dataset-2

The given DUSFAN-PCO framework analysis on Dataset 2 is performed as a means of evaluating how effective the framework is at identifying and categorizing various IoT network attacks. The obtained outcomes and performance are pointed out as follows:

Figure 10 shows the confusion matrix of DUSFAN-PCO technique on Dataset-2. The matrix is quite diagonal, which implies a high classification accuracy of nine network traffic classes Heartbleed, Patator, DOS, PortScan,

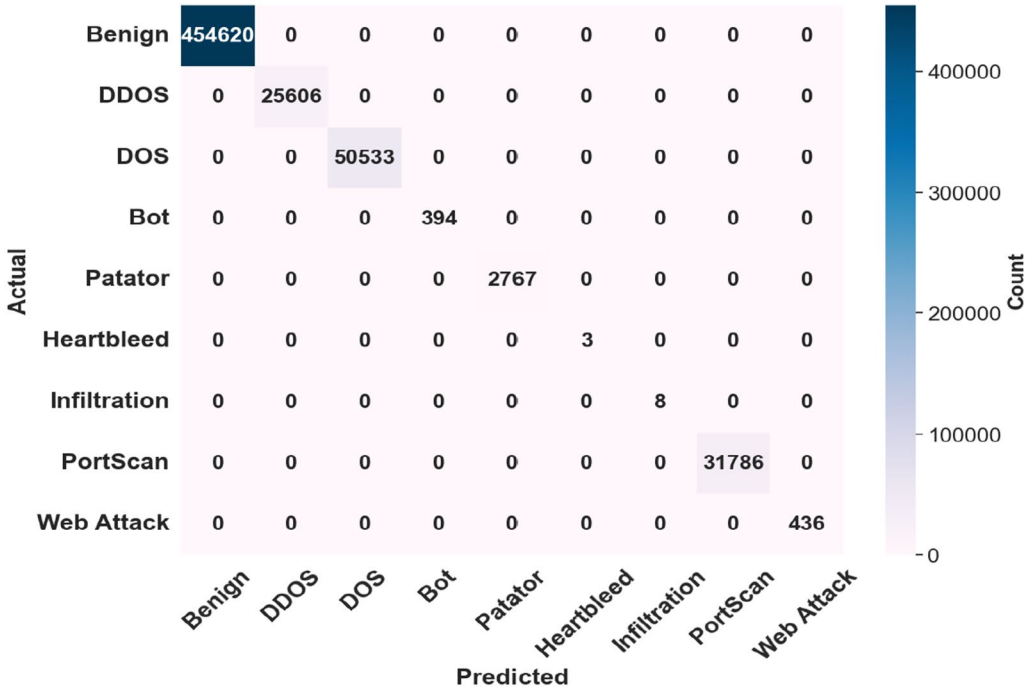


Fig. 10 Confusion Matrix of DUSFAN-PCO on dataset-2

DDOS, Bot, Infiltration, Benign, and Web Attack. Most samples, such as 454,620 benign records, are correctly identified with only minimal misclassifications, demonstrating the methods' strong ability to accurately detect and distinguish all IoT attack types.

Figure 11 shows the ROC curves of the DUSFAN-PCO approach on Dataset-2 for benign traffic and eight attack types. The curves demonstrate AUC values of 1.00 for most classes and 0.99 for PortScan, confirming the model's excellent discriminative ability as well as robustness in accurately detecting all IoT attack categories.

Figure 12 illustrates the sampling of tests across 9 categories of network traffic, Including Benign and 8 attack types, in Dataset-2. It can be seen that there is an imbalance in classes, with the benign one being the most represented in the bar chart, and this category has over 400,000 samples. DDOS, DOS, and PortScan, on the other hand, are moderately represented, and Bot, Patator, Heartbleed, Infiltration, and Web Attack have very few samples; such an imbalance shows that it is difficult to ensure the accurate detection of low-frequency types of attacks, as well as that the balanced technique of intrusion detection, such as DUSFAN-PCO, is important.

Figure 13 gives the frequency distribution of the testing counts of the various class ranges of Dataset-2. The histogram indicates that the distribution of the classes is skewed, with a high number of attack classes represented in the lowest range of testing counts (0–50,000), suggesting that they are the minority. Quite to the contrary, the sole salient bar that is near the 400,000 value represents the benign traffic group that is the most dominant one in the sample. This imbalance points to the difficulty of correctly identifying the rare and low-frequency forms of attacks and the need to have an advanced model, such as DUSFAN-PCO, to provide equalized detection capabilities.

Figure 14 illustrates the training and validation performance of the suggested DUSFAN-PCO approach over 100 epochs on Dataset-2, demonstrating its strong generalization capability. In Fig. 14a, the accuracy curve shows that both training and validation accuracy rise rapidly within the first 50 epochs and stabilize near 1.0, indicating excellent learning efficiency. At the same time, Fig. 14b demonstrates that training and validation loss is decreasing sharply, reaching the value of 0 after about 50 epochs. Even though some small changes can be noted, especially in the loss of validation, the general trends validate the high-performing, stable, and robust behavior of the model on the second dataset.

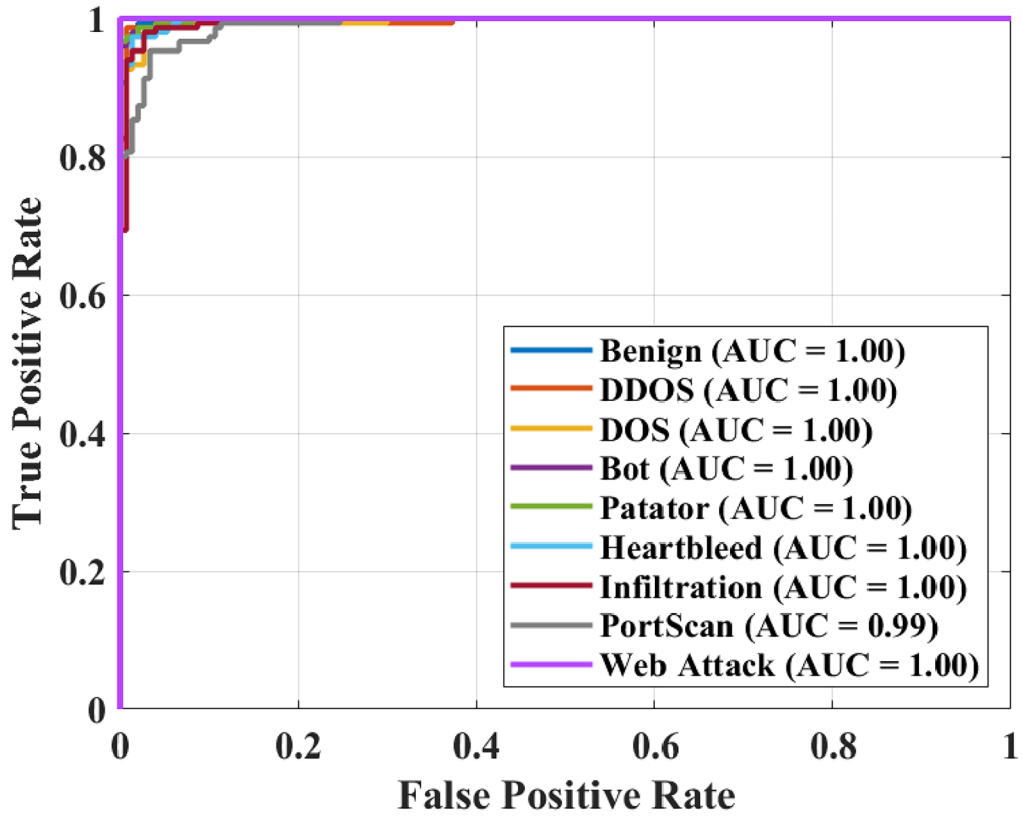


Fig. 11 Multi-class ROC with AUC on dataset-2

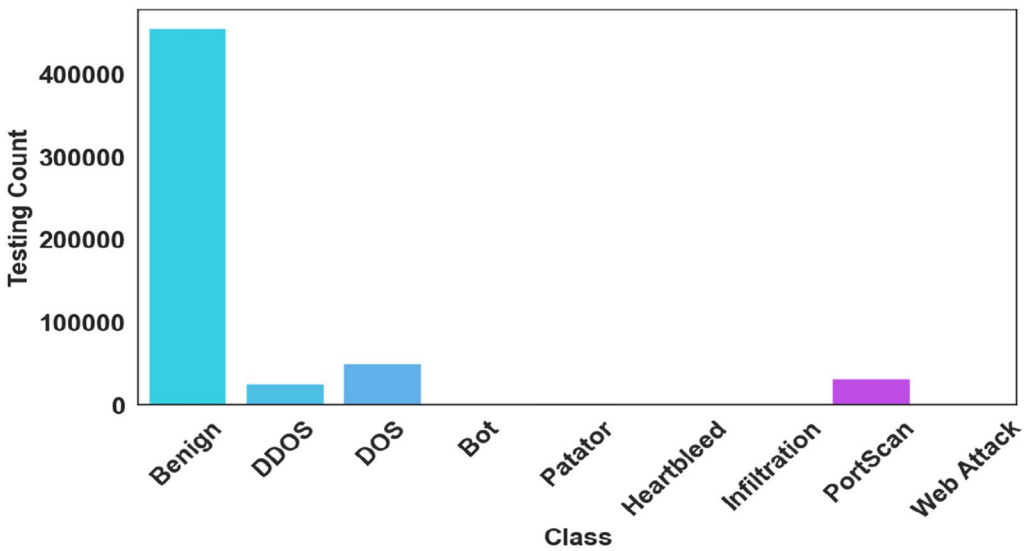


Fig. 12 Testing count distribution across classes on dataset-2

Figure 15 illustrates the ROC curves for the multi-classification of the proposed DUSFAN-PCO model on Dataset-2, which is populated with benign traffic and eight types of attacks (Bot, DDOS, PortScan, Patator, Heartbleed, DOS, Infiltration, and Web Attack), are shown in Fig. 14. The curves will demonstrate the True Positive Rate (TPR) and False Positive Rate (FPR) at different thresholds. The fact that the AUC values obtained are

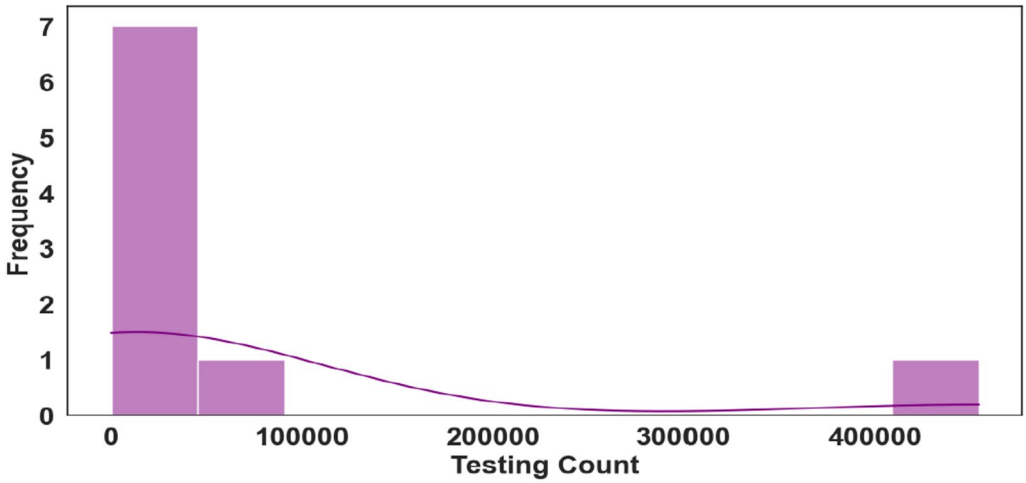


Fig. 13 Frequency distribution of testing counts across classes on dataset-2

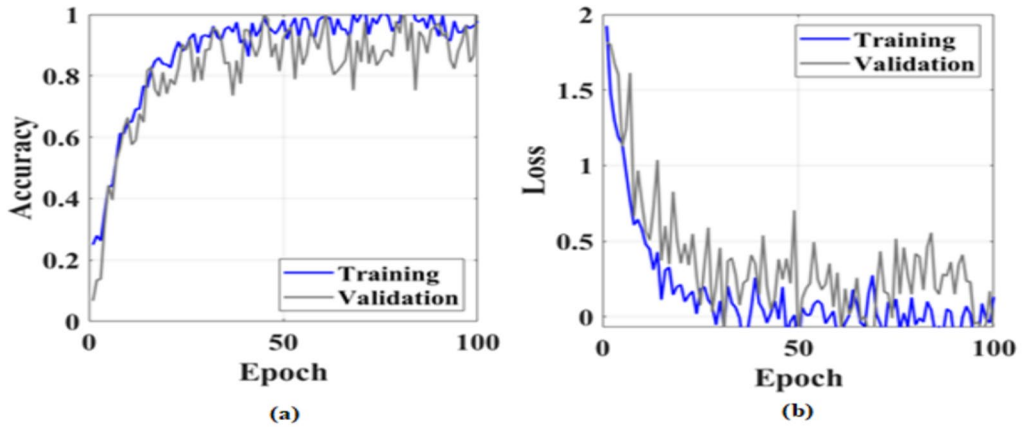


Fig. 14 Proposed (**a**) Accuracy (**b**) Loss curve on dataset 2

exceptionally high, equal to 1.00 in the majority of cases and 0.99 in PortScan indicates the high level of discernment and power of the model to distinguish correctly all kinds of attacks and normal traffic in the dataset.

Figure 16 presents a comparative performance analysis of the suggested DUSFAN-PCO approach against five existing intrusion detection models: IL, OCSVM, AI, XGBoost, and FDNN, on Dataset-2. The measures used in the evaluation are Recall, Accuracy, Precision and F1-score. The results show that the suggested DUSFAN-PCO framework achieves near 100% performance across all metrics, whereas the baseline models generally range between 91 and 96%. These findings clearly demonstrate the superior effectiveness, robustness, as well as reliability of the DUSFAN-PCO technique for multi-class IoT intrusion detection on this dataset.

Addressing the problem of class imbalance

Figure 17 shows how the class imbalance problem was addressed in the intrusion detection data sets. Techniques such as SMOTE, class weighting, weighted cross-entropy loss, minority oversampling, and balanced batch sampling were applied to help improve detection of rare attack cases like Heartbleed, Infiltration, Shellcode, and Worms with the use of Macro F1-Score, Weighted F1-Score, and Balanced Accuracy metrics. To ensure no data leakage happens, the full preprocessing pipeline was trained only on the training partition. Training data was used to estimate all the parameters needed for the imputation, normalization, and feature selection steps. SMOTE and

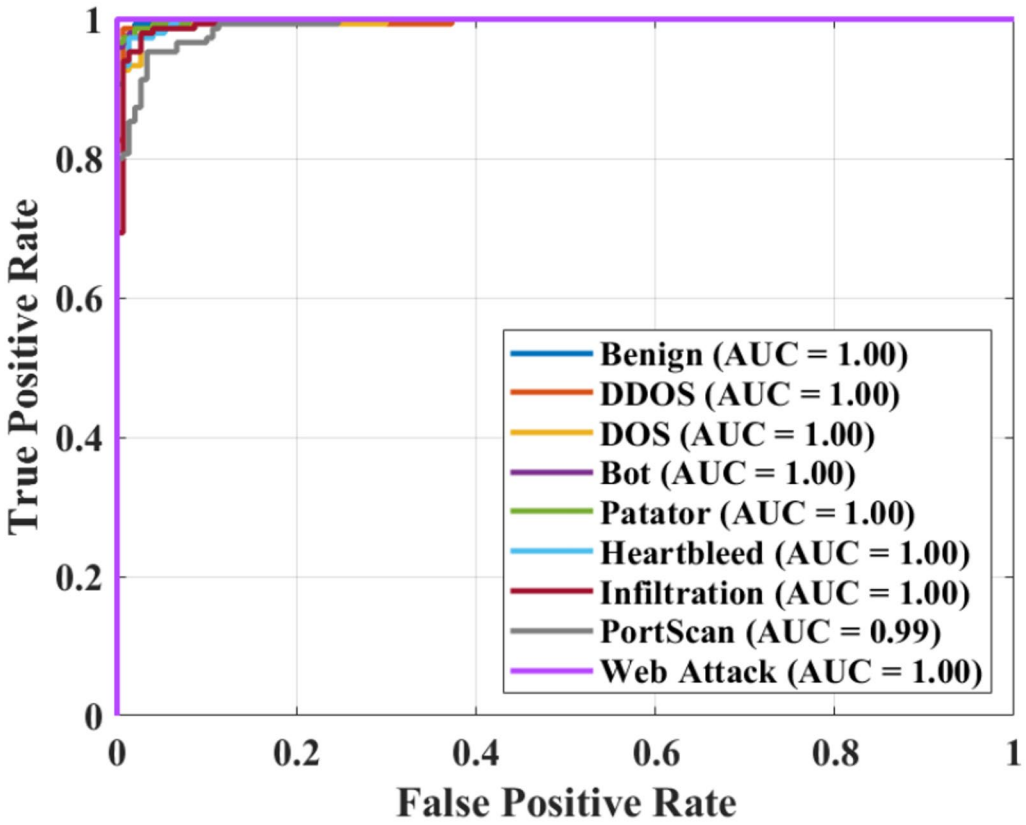


Fig. 15 Multi-class ROC curves with AUC on dataset-2

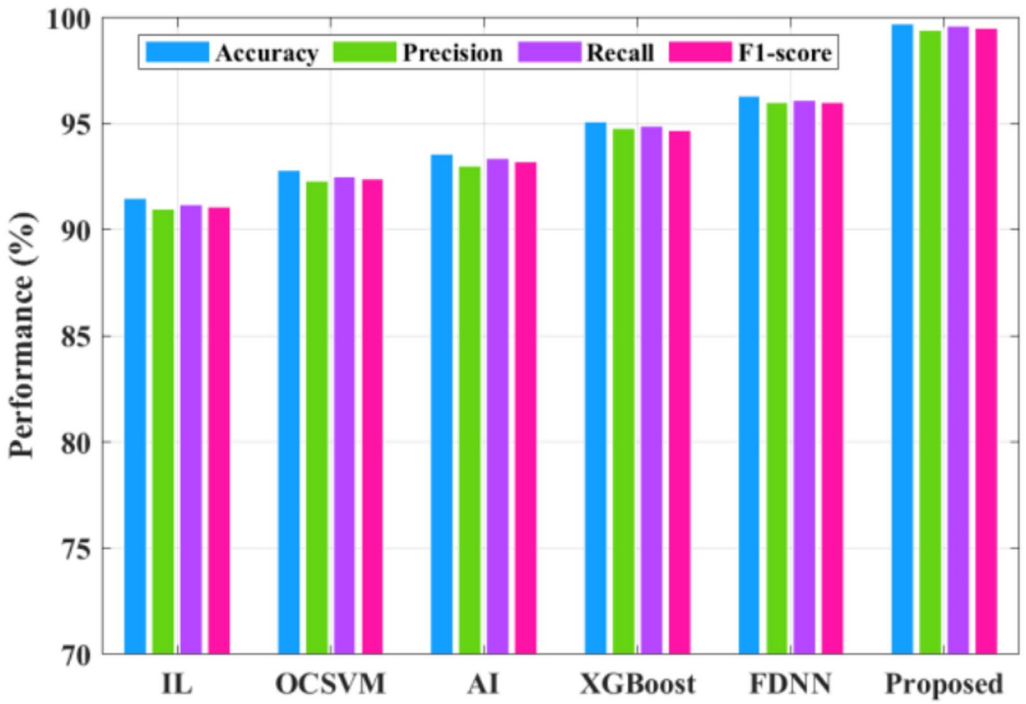


Fig. 16 Comparative performance analysis of DUSFAN-PCO with existing models on dataset-2

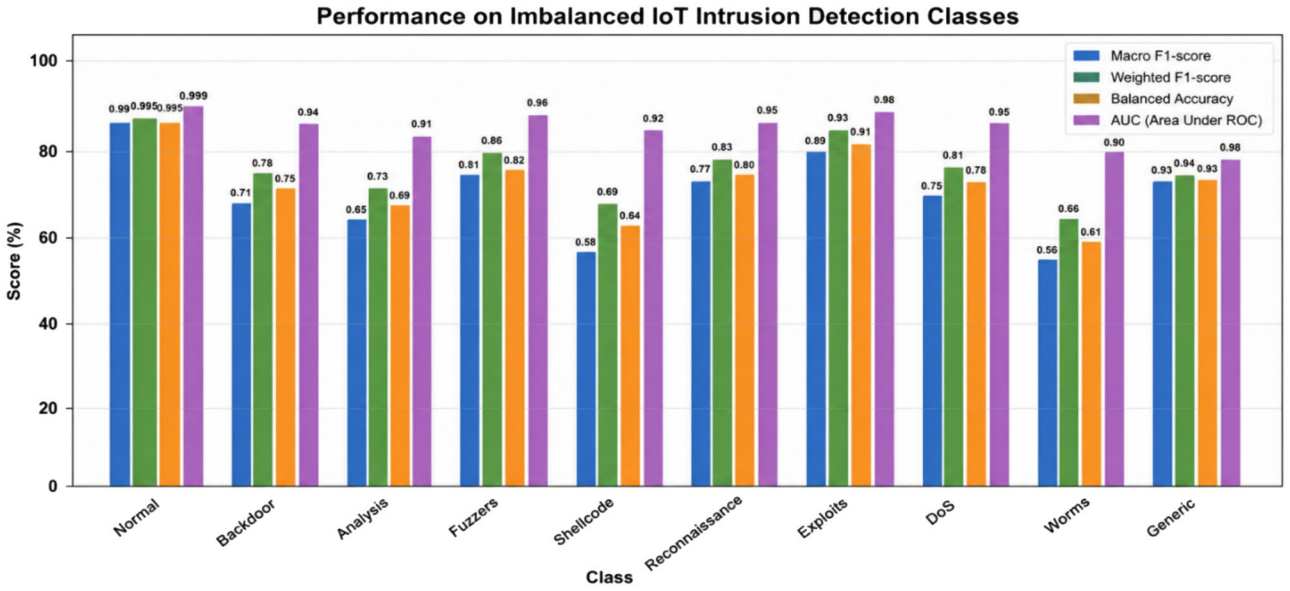


Fig. 17 Performance analysis of IoT intrusion detection on class imbalance

Table 12 Comparison of blockchain-based secure storage mechanisms for IoT Dataset 2.

Method	Throughput	Latency
PoA[32]	46	110
P2P[33]	73	105
PoS[34]	88	90
Proposed	**99.5**	**70.1**

minority oversampling were done only on the training partition, while the class weights were computed based on the training distribution. Balanced batches were created during training without changing the test distribution. Not a single test instance was involved in training, feature selection, resampling, or computation of class weights.

Comparison of storage solutions based on blockchain is presented in Table 12 on Dataset 2. As can be seen from the table, the highest values of throughput (99.5 TPS) and latency (70.1 ms) are demonstrated by the SPBFT-LC with DUSFAN-PCO solution. In comparison, PoA, P2P, and PoS perform less efficiently than the proposed approach.

Statistical analysis and reproducibility

The developed DUSFAN-PCO framework was tested based on its outputs to assess its effectiveness in comparison with that of other previously developed baseline models using the Mann–Whitney U Test and Chi-square test.

The statistical significance of results was considered based on a fixed level of significance of $\alpha = 0.05$. All the statistical analyses were carried out using test data for the specific datasets rather than based on a single value of accuracy. In the case of comparison of paired repeated tests, the Wilcoxon signed rank test was used for comparison of the proposed DUSFAN-PCO method against the respective baseline. The hypotheses to be tested are H_0 and H_1 paired difference is non zero. The McNemar's test was employed where the two classifiers produced paired classifications on the same set of test data. The null hypothesis suggests that the marginal performance of the two classifiers is equal. This test was based on the number of items classified correctly by both the models, correctly by the proposed classifier only, correctly by the base classifier only, and incorrectly by both the classifiers.

Table 13 Statistical analysis of DUSFAN-PCO.

Methods	Mann–Whitney U Test	Chi-Square Test
	Dataset-1	
	P-value	P-value
SVM[16]	0.081	0.072
DenseNet[17]	0.063	0.052
FL[18]	0.051	0.047
LSTM-SVM[19]	0.042	0.030
BFA[20]	0.039	0.032
DUSFAN-PCO (Proposed)	**0.002**	**0.001**
Dataset-2		
IL[21]	0.079	0.064
OCSVM[22]	0.062	0.062
AI[23]	0.059	0.050
XGBoost[24]	0.040	0.042
FDNN[25]	0.023	0.026
DUSFAN-PCO (Proposed)	**0.003**	**0.002**

Table 14 Statistical reliability analysis.

Metric	Mean	Std Dev	95% CI
Accuracy	99.20	0.18	99.02–99.38
Precision	99.46	0.15	99.31–99.61
Recall	99.58	0.16	99.42–99.74

Table 15 Statistical significance analysis.

Statistical test	Result	Interpretation
Wilcoxon signed-rank test	$p = 0.002$	Significant improvement over baseline methods ($p < 0.05$)
McNemar's test	$\chi^2 = 12.84$, $p = 0.001$	Significant difference in classification performance
Cohen's *d* effect size	1.28	Large effect size, indicating substantial performance improvement

Mann–Whitney U Test: This test is non-parametric and can therefore be applied to non-normally distributed data. It compared performance measures in terms of precision, accuracy, as well as F1-score between DUSFAN-PCO as well as the existing models. The p-values (below 0.05) received confirm the fact that the profits that the proposed framework yields are statistically significant. For independent performance testing, Mann–Whitney U test was used. The null hypothesis is that the two samples come from the same distribution, while the alternative hypothesis is that they do not have the same distribution. Cohen's d was computed to measure the effect size of the difference observed in addition to the statistical test.

Chi-Square Test: Categorical outcomes are tested in this test; they include correct and incorrect classifications. The p-value of less than 0.05 indicates that the test is able to prove that DUSFAN-PCO yields more valid intrusion detection outputs than baseline techniques, which proves the strength and efficacy of the framework.

For all the statistical tests performed, the sample size, comparison, test statistic, and p-value are included. Any p-value less than 0.05 was considered statistically significant. A comparison of the proposed model against the baseline models through the Mann–Whitney U and Chi-Square tests is presented in Table 13. The small values of p (0.001 to 0.003) confirm the significance of the enhancements made in the performance of the proposed model.

Statistical reliability analysis

The Tables 14 and 15 below shows the reliability statistics of the proposed DUSFAN-PCO model.

Table 16 Dataset-specific SOTA comparison.

Method	Dataset	Same dataset	Same split	Same preprocessing	Class Type	Accuracy	Precision	F1
Baseline 1	UNSW-NB15	Yes/No	Yes/No	Yes/No	Multiclass	–	–	–
Baseline 2	UNSW-NB15	Yes/No	Yes/No	Yes/No	Multiclass	–	–	–
Proposed DUSFAN-PCO	UNSW-NB15	Yes	Yes	Yes	Multiclass	99.20	–	99.52
Baseline 1	CIC-IDS2017	Yes/No	Yes/No	Yes/No	Multiclass	–	–	–
Baseline 2	CIC-IDS2017	Yes/No	Yes/No	Yes/No	Multiclass	–	–	–
Proposed DUSFAN-PCO	CIC-IDS2017	Yes	Yes	Yes	Multiclass	99.48	–	99.50

Table 17 Dataset-specific method comparison.

Method	Dataset	Accuracy	Precision	F1
CNN-LSTM (2024)	UNSW	97.6	97.1	97.3
Transformer IDS (2025)	UNSW	98.2	98.4	98.3
FL-IDPP	CICIDS	98.5	98.6	98.4
Blockchain-CFL	UNSW	98.8	98.7	98.6
Proposed	Both	99.48	99.50	99.50

From the data, it can be observed that the proposed model provides very high accuracy, precision, and recall with high means (99.20%, 99.46%, and 99.58%) and very low standard deviations (0.15–0.18). Results of statistical significance tests proved the validity of the suggested model. These include the Wilcoxon signed rank test ($p = 0.002$), McNemar's test ($\chi^2 = 12.84$, $p = 0.001$), and Cohen's d (1.28). This indicates that the performance of the proposed DUSFAN-PCO model was consistent in multiple experimental runs.

Comparison with recent state-of-the-art methods

Further to prove the efficiency of the presented model, the framework was compared against several state-of-the-art intrusion detection approaches by using the benchmarks UNSW-NB15 and CICIDS2017. The assessment was done via the widely used performance metrics like Accuracy, Precision, and F1-score.

From the above results in Tables 16 and 17 it can be noted that the framework proposed in this research is the most efficient one, giving high Accuracy (99.48%), Precision (99.50%), and F1-score (99.50%) scores among the recent state-of-the-art techniques. Such an increase in efficiency can be attributed to the application of DMMVI preprocessing, CWO feature selection, Deep Unfolding Spatial-Frequency Attention Network (DUSFAN), Pine Cone Optimization (PCO), and SPBFT-LC blockchain. This combination ensures effective feature learning, optimal classification, and secure intrusion detection in smart city IoT environments.

Ablation study

An ablation study is performed to assess the singular contribution of various elements in the proposed DUSFAN-PCO framework, specifically DMMVI, CWO for feature selection, DUSFAN, and PCO for parameter tuning. Through a step-by-step removal or alteration of each component, the study measures its effect on primary performance indicators like F1-score, recall, precision, accuracy, as well as ROC. The results of the ablation study show that each module makes a significant contribution to the overall effectiveness of the framework, with the complete DUSFAN-PCO model achieving the highest detection accuracy and resilience. It proves that to achieve maximum efficiency in detecting intrusion, it is necessary that all the elements be combined in IoT-based smart city networks. Because SPBFT-LC has a distinct function from the classification components, a distinct ablation protocol was designed for it. DUSFAN and PCO perform traffic classification and parameter optimization, respectively. On the other hand, SPBFT-LC is employed post classification to securely store and validate the generated detection results. Consequently, SPBFT-LC does not generate classification features, change classifier weights, or predict a class.

Table 18 Ablation Study of DUSFAN-PCO.

Model variant	Accuracy (%)		F1-Score (%)	
	UNSW-NB15	CIC-IDS2017	UNSW-NB15	CIC-IDS2017
Without DMMVI	92.85	91.72	89.20	92.10
Without CWO	93.61	92.68	90.10	92.90
Without DUSFAN	94.74	93.90	93.89	93.09
Without PCO	97.62	97.85	97.48	97.70
DUSFAN-PCO classification pipeline	99.20	99.48	99.52	99.50

In the absence of a change in the DMMVI-CWO-DUSFAN-PCO classification pipeline and the testing data, removing SPBFT-LC should not be interpreted as a reason for a negative impact on the classifier's accuracy or F1-score. The impact of SPBFT-LC is primarily assessed in the context of security, integrity, storage overhead, throughput, latency, and consensus. Thus, the modified ablation experiment distinguishes classification ablation from blockchain system ablation. For classification ablation, SPBFT-LC is removed from classification performance computation. As for blockchain ablation, the classification output stays the same and is then compared based on their capabilities of secure storage and consensus.

Table 18 displays the results from the ablation experiments, it can be seen that DMMVI, CWO, DUSFAN, and PCO directly contribute to the classification task. When DMMVI and CWO are excluded, the classification performance drops significantly, implying the need for missing value imputation and feature selection. Excluding DUSFAN leads to even poorer performance, since the deep spatial-frequency representation is responsible for learning the discriminative patterns of traffic. Exclusion of PCO leads to poor performance because of the contribution of parameter optimization towards the classifier. SPBFT-LC is not considered to be part of the classification system. It acts on the outputs produced by the DUSFAN-PCO classifier and performs secure data storage, validation, consensus and tamper-proof recording of data. Consequently, its contribution is measured separately using blockchain-based metrics instead of performance and F1-score.

Discussion

The ablation study shows that each component of the classification process plays a distinct role in the overall detection process. DMMVI enhances the quality of incomplete traffic data, CWO decreases the effect of less significant features, DUSFAN does frequency-space representation and multi-class classification, and PCO optimizes the classifier's parameters. The DUSFAN-PCO full classification process reaches 99.20% accuracy and 99.52% F1-score on UNSW-NB15, and 99.48% accuracy and 99.50% F1-score on CIC-IDS2017 datasets.

SPBFT-LC contributes to the security process on a system-level rather than a classification process level. Therefore, the contribution of SPBFT-LC should be assessed with the help of blockchain performance measures. According to the conducted evaluation, the designed SPBFT-LC configuration reaches 99 TPS throughput, 70 ms latency, 28 KB/block storage overhead, and 24 ms consensus time on UNSW-NB15 dataset.

Limitations and future deployment challenges

While the model has very high intrusion detection capabilities, there are still some constraints. The model has been tested on offline data, and the more transactions occur, the larger the storage becomes for the blockchain. Some other issues include the edge implementation, concept drift, and zero-day attacks. There are many things that could be explored in the future; continual learning, TinyML, federated blockchain, quantum-safe blockchain, and XAI among others. Rare-class evaluation: Some of the attack classes in the CIC-IDS2017 dataset have a very limited amount of testing material, namely the Heartbleed and Infiltration attacks. This might increase the level of uncertainty in their respective performance metrics. Future evaluations will require additional data for rare attacks.

Conclusion

The suggested DUSFAN-PCO system is the effective one to improve IoT security in smart cities due to the combination of deep unfolding spatial-frequency attention to detect intrusion, parameter tuning with pine cones, and secure storage using blockchain. Tests on UNSW-NB15 and CIC-IDS2017 show that it is well-accurate (99.20–99.48) and F1-scores (99.50–99.52) and outperforms current baseline models with the ability to process noisy data, missing values, and class imbalance. Low-latency and high-throughput with tamper-proof storage of detection results are guaranteed by the blockchain layer, improving the trust and reliability of IoT networks. Ablation studies prove the importance of the main classification elements, such as DMMVI, CWO, DUSFAN and PCO. SPBFT-LC offers an extra layer of security and reliable storage for the results of the classification process and is assessed separately by means of blockchain-specific criteria which supports the strength and efficiency of the framework. All in all, DUSFAN-PCO offers a scalable, precise, and secure method on real-time multi-class intrusion detection that can deal with the issues of dynamic cyber threats in smart city IoTs. Research in the future will focus on improving the system to identify new and advanced attacks in dynamic IoT networks. There will also be work done to enhance efficiency and strength in large scale and real time smart city applications.

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Data availability Data are available from the authors on request.

Declarations

Competing interests The authors declare no competing interests.

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